

Teacher Guide — Electrostatics (Electric charge, Coulomb's law, electric field, Gauss's law)

Physics | Grade 11-12

Period 1

ENTRY TICKET

You have two small identical metal spheres, A has $+3e$ and B has $-1e$ ($e = 1.60 \times 10^{-19} \text{ C}$). You touch them together and then separate them. What is the net charge on A and B after separation, in units of e ? Briefly explain one sentence why.

Expected response: Net total charge = $+2e$. After touching, charge redistributes between the two identical spheres so each ends with $+1e$ on average, or one way answer: each sphere has $+1e$ (so $A = +1e$, $B = +1e$).

Reason: charge is conserved and redistributes when conductors touch.

Teacher Script

0-5 min: Warm-up and surface prior ideas

Say: "Read the ticket. Put a short answer on the board: what are the charges on A and B after they touch? One line explanation." Walk quickly while students write. After 90 seconds, ask one student with an answer to read theirs. If a student says 'charges disappear' or 'they cancel and go to zero' ask: 'What do you mean by disappear? Where did the electrons go?' Collect 2–3 different answers aloud. Say: 'We'll test that idea today.'

Board: Write two columns labelled 'Answer from students' and 'Why?'. Copy one typical wrong answer: 'charges destroyed' and one correct: 'redistributed, conserved'.

- Anticipate: If they neutralise, how can the total stay the same?
- Anticipate: Can one sphere end up with more than $+3e$ after touching?
- Anticipate: Do we have to count electrons only, or protons too?

5-13 min: Demonstration and measurement (rubbing and counting idea)

Say: "I'll show a quick demo with two pith-ball-like spheres drawn here and a cloth. When I rub the cloth on the right sphere it gains electrons; the left is left neutral." Point to diagram. Ask: 'If the rubbed sphere gains $-5e$, what is its charge in coulombs? Use $e = 1.60 \times 10^{-19} \text{ C}$.' Wait for a volunteer to compute: $-5 \times 1.60 \times 10^{-19} \text{ C} = -8.0 \times 10^{-19} \text{ C}$. Write the arithmetic step as you say it. Then say: 'Now if those two spheres touch and are identical conductors, charge spreads evenly. What happens to the number of elementary charges? Does the total number change?' Pause for answers and emphasise: 'Total number of excess electrons stays the same — conservation.'

Board: Draw two circles labelled S1 (rubbed) and S2 (neutral). Next to S1 write ' $-5e$ ' and compute ' $-5 \times 1.60 \times 10^{-19} \text{ C} = -8.0 \times 10^{-19} \text{ C}$ ' showing the multiplication and unit 'C'. Arrow between spheres showing 'touch -> redistribute'.

- Anticipate: Why does it spread evenly? (Because they're identical conductors and charges

move until equilibrium.)

- Anticipate: Why do we use e and not count protons? (Free charge transfer is electrons; protons stay in nuclei.)
- Anticipate: What if one sphere is bigger? (Charge divides in proportion to capacity — beyond today's scope.)

13-23 min: Derive $q = n e$ from counting electrons

Say: "We can describe any net free charge as a number of elementary charges times e . Write the relation with me: $q = n e$. Here q is the measured charge in coulombs, n is an integer (can be negative), and $e = 1.60 \times 10^{-19} \text{ C}$. Example: if $2.0 \times 10^{-18} \text{ C}$ of extra negative charge is measured, what is n ?" Wait for calculation: $n = q/e = (-2.0 \times 10^{-18} \text{ C}) / (1.60 \times 10^{-19} \text{ C}) = -12.5$. Point out: " n must be an integer for an isolated object with free charges. So a measured q like $-2.0 \times 10^{-18} \text{ C}$ would mean about -12 or -13 electrons extra; the exact integer depends on measurement precision. For classroom problems we keep numbers that divide cleanly." Emphasise units each step: 'C divided by C gives unitless n .'

Board: Write: ' $q = n e$ ' then beneath: ' $q \text{ (C)} \div e \text{ (C)} = n \text{ (unitless integer)}$ '. Show the sample division: $(-2.0 \times 10^{-18} \text{ C}) \div (1.60 \times 10^{-19} \text{ C}) = -12.5 \rightarrow$ approximately -12 or -13 electrons depending on rounding/precision.

- Anticipate: But you said n must be an integer — so how can we get -12.5 ? (Answer: measurement precision; real objects have integer n , measurement rounding gives non-integer.)
- Anticipate: Can n be positive? (Yes: positive n means missing electrons, net positive charge.)
- Anticipate: Why is e the same for all electrons? (Elementary charge is the magnitude of electron/proton charge.)

23-31 min: Guided practice: convert and check conservation

Hand out or project three quick problems and solve the first together aloud: 1) A rod has $+4e$. What is q in C? 2) A particle has $q = -3.2 \times 10^{-19} \text{ C}$. How many elementary charges is that? 3) Two spheres, one with $+6e$ one with $-2e$ touch — what are final charges on each if identical? For (1) say: $q = n e = 4 \times 1.60 \times 10^{-19} \text{ C} = 6.4 \times 10^{-19} \text{ C}$. Write each step and the unit. For (2) compute $n = q/e = -3.2 \times 10^{-19} / 1.60 \times 10^{-19} = -2 \rightarrow$ answer: $-2e$. For (3) total = $+4e$; after touching each has $+2e$. Ask students to do these in pairs for 4 minutes and write answers; circulate and correct work.

Board: Write the three problems and solve #1 and #2 step-by-step with units. For #3 write: ' $+6e + (-2e) = +4e \rightarrow$ each sphere $\rightarrow +2e$ '.

- Anticipate: Why did we divide by e to get n ? (Because $q = n e$, so $n = q/e$.)
- Anticipate: If q is very small, can n be zero? (Yes; zero net excess charge means $n = 0$.)
- Anticipate: What if q is not a multiple of e exactly? (Measurement/rounding; physical n is integer.)

31-37 min: Formative check — quick quiz and misconceptions

Give one quick written question: 'A neutral conductor touches a rod with $-8e$. After separation, what is the charge on the conductor (in C)? Explain in one sentence how conservation works here.' Collect answers quickly or have students hold up a card with their answer. If many show '0' or 'disappeared', say: 'No — charges aren't destroyed. The conductor gained $-8e$ overall; that's $-8 \times 1.60 \times 10^{-19} \text{ C} = -1.28 \times 10^{-18} \text{ C}$.' Reiterate: 'When opposite charges meet they neutralise locally, but the total charge of the isolated system is conserved.'

Board: Write the quick question, then the correct arithmetic: ' $-8e = -8 \times 1.60 \times 10^{-19} \text{ C} = -1.28 \times 10^{-18} \text{ C}$ ' and

one sentence: 'Conservation: electrons moved from rod to conductor; total charge of system unchanged.'

- Anticipate: If opposite charges cancel, why don't they disappear from the universe? (They combine to make neutral atoms but the electrons and protons remain; net charge is conserved.)
- Anticipate: When they neutralise is energy released? (Yes, sometimes as light/heat — not required today.)

37-40 min: Consolidation and homework

Summarise aloud: 'We saw charge is $q = n e$ with $e = 1.60 \times 10^{-19} \text{ C}$, and we must keep units. Touching conductors redistributes charge but the total number of excess elementary charges stays the same. For homework: three problems — convert between n and q , and one touching conductors problem similar to today's. Due next lesson.' Hand out the sheet or write assignment on board.

Board: Write 'Homework: 3 problems — conversions and one touching conductors. Due next lesson.'

- Anticipate: Can we use 1.6×10^{-19} for e instead of 1.60×10^{-19} ? (Yes, but keep consistent significant figures.)
- Anticipate: Will we need Coulomb's law today? (No — it's listed for the course but not used in calculations today.)

Blackboard Notes

- Quantisation of Charge
- Example Calculations
- Conservation when conductors touch
- Homework
- $q = n e$ (q in coulombs, n is integer, $e = 1.60 \times 10^{-19} \text{ C}$)
- Units: $q \text{ (C)} = n \text{ (unitless)} \times e \text{ (C)}$
- Example 1: $+4e \rightarrow q = 4 \times 1.60 \times 10^{-19} \text{ C} = 6.4 \times 10^{-19} \text{ C}$
- Example 2: $q = -3.2 \times 10^{-19} \text{ C} \rightarrow n = q/e = -3.2 \times 10^{-19} / 1.60 \times 10^{-19} = -2 \rightarrow -2e$
- Touching identical conductors: charges add and redistribute but total n conserved
- Example touching: $+6e$ and $-2e \rightarrow$ total $+4e \rightarrow$ each sphere (identical) ends with $+2e$
- Conservation note: 'charges destroyed' is wrong — charge is transferred; net charge of isolated system is constant
- Diagram: Two circles $S1$ and $S2$, label $S1$: $-5e$, arrow 'touch' to $S2$, then show separated both with $-2.5e$ (explain rounding) or state 'redistribute evenly $\rightarrow$ each -2 or $-3e$ depending on integer constraints'
- Diagram: Two identical spheres labelled $+6e$ and $-2e$, arrow 'touch' then separated both $+2e$
- Formula: $q = n e$
- Formula: $F = k \{q_1 q_2 \over r^2\} = \{1 \over 4\pi \epsilon_0\} \{q_1 q_2 \over r^2\}$

Activities

Counting Charge on a Charged Balloon (10 min, experiment)

Materials: 2 round balloons (one per pair), wool or nylon cloth (a small square of fabric per pair), small scraps of paper (about 10 per pair, $\sim 1 \text{ cm}$ squares), metric ruler (class set or one per pair), calculator (one per student or pair), printed data slip (one per pair) with value $e = 1.60 \times 10^{-19} \text{ C}$ and space for calculations, stopwatch or phone timer (one per pair)

•

- 1) Organize students into pairs and give each pair the materials: 1 balloon, fabric, ~10 paper scraps, ruler, calculator, and one printed data slip. Tell them: 'You will charge a balloon, use it to lift papers, count how many elementary charges correspond to that measured charge, and show the derivation step by step.'
- 2) Demonstrate at the front: inflate a balloon (not fully; ~20 cm diameter), tie it, rub it with the fabric for 30 seconds, and bring it near a loose paper scrap to show attraction. Say: 'We measure how many papers are attracted; each attracted paper indicates a minimum net charge magnitude that can hold that paper against gravity for a short time.'
- 3) Start the 10-minute timer and instruct pairs to run the procedure on their slip: rub the balloon 30 s, bring it near the pile, count how many papers stick, measure the shortest distance at which the paper is attracted using the ruler, and record times and counts. While they work, circulate and check that pairs rub the balloon for 30 s, count visible attracted papers (not ones moved by air), and write numbers with units.
- 4) After data collection (about 5–6 minutes), instruct pairs to compute a lower-bound estimate of the balloon's net charge: (a) assume each paper has mass $m \sim 0.02 \text{ g}$ (teacher-provided on slip) and that electrostatic attraction force at closest distance r must be at least equal to paper weight mg ; (b) use Coulomb's law form $F = k Q q / r^2$ and treat q as one paper's induced charge, approximate $q = Q$ (paper area / balloon area) is too detailed—instead derive a simple lower bound by assuming $q \sim Q/10$ (explain this is conservative); (c) solve for Q in coulombs, then divide by $e = 1.60 \times 10^{-19} \text{ C}$ to get n . Pairs must show each algebra step and units on the slip.
- 5) End class by collecting one example slip from a pair and ask three pairs to read their final n value aloud. While students report, note any showing disappearance-of-charge misconception and correct it: emphasize conservation and redistribution of charge when objects touch, not destruction.

Student instructions:

- 1) Inflate and tie your balloon to about 20 cm diameter. Start the stopwatch and rub the balloon with the fabric for 30 seconds. Write the rubbing time on your slip.
- 2) Place ~10 paper scraps on the desk. After rubbing, slowly bring the balloon toward the top paper and count how many papers are attracted and stick to the balloon. Record the count N_{papers} (unit: count). Do not let air currents move the papers.
- 3) Measure the closest distance r (in meters) from the balloon surface to the paper when attraction first occurs; record r with units (use cm and convert to m on your calculator). Record the mass per paper $m = 0.02 \text{ g} = 2.0 \times 10^{-5} \text{ kg}$ (given).
- 4) Estimate a conservative lower bound for the balloon's net charge Q (in coulombs) as follows, showing every algebra step and units: (a) write the paper weight $W = m g$, with $g = 9.8 \text{ m/s}^2$ and units N; (b) assume the electric force on one paper is $F_e \geq W$; (c) use Coulomb's law $F_e = k |Q q| / r^2$ with $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$; (d) to avoid measuring q , assume $q \sim Q/10$ (a conservative redistribution assumption) and substitute to get $F_e \sim k Q (Q/10) / r^2 = k Q^2 / (10 r^2)$; (e) set $k Q^2 / (10 r^2) = W$ and solve step-by-step for Q (show algebra and units); (f) compute numerical Q in C, then compute $n = Q / e$ and give n as an order-of-magnitude integer (include units: coulombs and number of elementary charges).
- 5) Write the final answers on the slip: rubbing time (s), N_{papers} (count), r (m), Q (C), and n (unitless count of e). Be ready to explain one algebra step to the teacher.

Success looks like: Each pair has a completed slip showing rubbing time (s), paper count (unit: count), measured r with units (m), and numerical values for Q in coulombs and n (unitless) with algebra steps visible.; While

circulating you can see at least one pair carrying out the algebra on their calculator or paper with units carried through each step (e.g., $\text{kg}\cdot\text{m}/\text{s}^2$ for weight, $\text{N}\cdot\text{m}^2/\text{C}^2$ for k).; At least two students can point to the line on their slip where they set electric force $\geq$ weight ($F_e \geq mg$) and used Coulomb's law to solve for Q .

Support: If a pair cannot start, pair them with a stronger partner and give this worked first item on the slip:

Example work: $m = 2.0 \times 10^{-5} \text{ kg}$, $g = 9.8 \text{ m/s}^2$ so $W = m g = 2.0 \times 10^{-5} \text{ kg} \times 9.8 \text{ m/s}^2 = 1.96 \times 10^{-4} \text{ N}$. Use $r = 0.02 \text{ m}$, $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, assume $q = Q/10$ so $k Q^2/(10 r^2) = W$. Rearranged: $Q^2 = 10 W r^2 / k$. Plug numbers: $Q^2 = 10 \times (1.96 \times 10^{-4} \text{ N}) \times (0.02 \text{ m})^2 / (8.99 \times 10^9) =$ compute this on calculator $= 8.71 \times 10^{-21} \text{ C}^2$ so $Q = 9.34 \times 10^{-11} \text{ C}$. Then $n = Q / e = 9.34 \times 10^{-11} \text{ C} / 1.60 \times 10^{-19} \text{ C} = 5.8 \times 10^8$ (about 6×10^8 elementary charges). Use this as a template and replace numbers with your measured r and paper count.

Extension: For pairs who finish early: instead of assuming $q = Q/10$, estimate q by modelling the paper as a disk of area $A_{\text{paper}} = 1 \text{ cm}^2$ and the balloon as a sphere of radius $R \sim 0.10 \text{ m}$. Derive $q \sim Q (A_{\text{paper}} / (4\pi R^2))$.

Recompute Q and n using this area-based relation, show the algebra with units, and compare the result to the conservative $q = Q/10$ assumption; explain which gives a larger n and why.

Checkpoint Questions

- [apply] A plastic rod gains $6.4 \times 10^{-18} \text{ C}$ of net charge after being rubbed. Using $q = n e$ with $e = 1.60 \times 10^{-19} \text{ C}$, compute n (the number of elementary charges) and state whether electrons were gained or lost.

Expected: $n = q / e = (6.4 \times 10^{-18} \text{ C}) / (1.60 \times 10^{-19} \text{ C}) = 40$. Positive q means a deficit of electrons (net +40 e), so 40 protons' worth of charge; equivalently the rod lost 40 electrons. Units: dimensionless $n = 40$.

- [apply] Two small metal spheres are isolated from everything else. Sphere A has charge $+3e$ and sphere B has charge $-2e$. If they touch and then are separated, what is the charge on each sphere in coulombs? Show arithmetic with units.

Expected: Total charge $= (+3e) + (-2e) = +1e$. When they touch and are identical spheres, charge shares equally: each ends with $+1e/2$. In coulombs: $e = 1.60 \times 10^{-19} \text{ C}$, so each has $(0.5 \times 1.60 \times 10^{-19} \text{ C}) = 8.0 \times 10^{-20} \text{ C}$. Units shown: C.

- [apply] Estimate the number of electrons that carry a charge of 1.0 microcoulomb ($1.0 \times 10^{-6} \text{ C}$). Give your calculation and an order-of-magnitude comment.

Expected: $n = q / e = (1.0 \times 10^{-6} \text{ C}) / (1.60 \times 10^{-19} \text{ C}) \sim 6.25 \times 10^{12}$ electrons. Order of magnitude: about 10^{13} electrons. Units: n is dimensionless count.

EXIT TICKET

A small paper square is given a net charge of $-2.4 \times 10^{-17} \text{ C}$. (a) How many extra electrons does it carry? (b) Write the result both as an integer number of electrons and as the same charge in units of e . Show units and brief working.

Success indicator: Student calculates $n = q/e = (-2.4 \times 10^{-17} \text{ C})/(-1.60 \times 10^{-19} \text{ C}) = 150$ electrons, states the sign (extra electrons: 150), and shows the charge as $-150 e$ or $-150 \times e = -2.4 \times 10^{-17} \text{ C}$ with units.

Homework

- Convert each charge to a number of elementary charges and state the sign (extra electrons or deficit): (a) $+4.8 \times 10^{-19} \text{ C}$, (b) $-8.0 \times 10^{-18} \text{ C}$, (c) $+1.0 \times 10^{-6} \text{ C}$. Show work with $q = n e$ and include units.
- Two identical metal spheres carry charges of $+8e$ and $+2e$. They touch and are separated. Calculate the final charge on each sphere in elementary charges and in coulombs.
- Short written explanation (3–5 sentences): Explain why charge appears to be 'lost' when an

object is neutralised by contact, and correct the misconception by stating what actually happens to electrons and the net charge of the isolated system.

Estimated time: 30 min

MENTOR MOMENT — THE COIN AND THE COMB

A student once rubbed a plastic comb on their sleeve, then noticed it picked up tiny paper bits. They counted this as 'magic' until we calculated the comb carried about 10^{12} extra electrons — a huge number compared with atoms but a tiny fraction of the comb's electrons.

Takeaway: Even a small measurable charge corresponds to trillions of electrons moving; that helps make abstract charges feel concrete when you attach numbers and units.

Period 2

ENTRY TICKET

Write a short answer (2 lines max): A small plastic rod was rubbed with wool and now has a net charge of $+3e$. What is that charge in coulombs? Show one calculation step and give the number with units. If you think it is impossible to answer, write why.

Expected response: Calculation: $q = 3 \times (1.60 \times 10^{-19} \text{ C}) = 4.8 \times 10^{-19} \text{ C}$. Answer: $4.8 \times 10^{-19} \text{ C}$. If a student writes 'impossible' they should say they don't know the value of e .

Teacher Script

0-5 min: Activate prior knowledge: check quantisation

Read the entry ticket aloud: 'A rod has charge $+3e$. What is that in coulombs?' Wait 30 seconds for students to finish. Ask one student to read their calculation step and number. If a student writes 'impossible', ask: 'What number would let us compute this? (Expect: value of e .)' Repeat if necessary: point to the printed constant on the sheet: $e = 1.60 \times 10^{-19} \text{ C}$. Tell class: 'Today we will use that relation again; we must always write units.'

Board: Write the example students did: 'Example: $+3e \rightarrow ? \text{ C}$ ' then write ' $e = 1.60 \times 10^{-19} \text{ C}$ ' and the single worked calculation line ' $3 \times 1.60 \times 10^{-19} \text{ C} = 4.8 \times 10^{-19} \text{ C}$ '.

- Anticipate: Why is e written with a lowercase letter? (Because e is the magnitude of the electron charge; it's a defined constant.)
- Anticipate: Can n be negative? (Yes — n is negative if the net charge has more electrons than protons.)
- Anticipate: Do we need more than 2 significant figures for e ? (Use $1.60 \times 10^{-19} \text{ C}$ here; we'll keep that precision.)

5-17 min: Introduce conductors vs insulators

Say: 'Hold the metal water bottle in your hand.' Pause. 'Metals are conductors: electrons can move freely through them.' Point to a plastic ruler and say: 'Plastic is an insulator: charges we put on it stay where we put them.' Draw a quick physical example: a metal sphere and a plastic rod. Ask: 'If I touch a charged metal sphere, where do the charges go?' Let students answer. If no answer: 'They spread over the surface.' Then demonstrate with a picture: mark $+$ charges spreading over the outer surface of the sphere and on the rod mark a few $+$ charges in one spot to show localization. Emphasize units: 'Charge measured in coulombs.' Finally ask: 'Which would let charges move to the outside when charged, conductor or insulator?' Confirm: conductor.

Board: Draw two labeled sketches: a circle labeled 'metal (conductor)' with $+$ signs spread on outer surface; a rod labeled 'plastic (insulator)' with $+$ signs clumped. Write two short bullets: 'Conductor: charges move and redistribute (surface),' and 'Insulator: charges stay where placed (localized).'

- Anticipate: Does every metal conduct equally well? (No — conductivity differs, but all conductors allow charge to move compared to insulators.)
- Anticipate: What about semiconductors? (Not covered today; they are in between.)
- Anticipate: Do neutral conductors have charges on the surface? (If net charge is zero, there is no excess on the surface; only when charged do we get excess on the surface.)

17-27 min: Operational definition: electric field

Say: 'Now we define electric field by an experiment.' Place an imagined small positive test charge q_t at a point P near a source charge $+Q$. Ask: 'What do we measure there?' Guide students to say 'a force on the test charge.' State the operational definition exactly: 'The electric field at a point is the force experienced by a unit positive test charge placed there (without disturbing the source charges); equivalently, force per unit test charge.' Now place numbers: 'If a $2.0\ \mu\text{C}$ test charge experiences a force of $6.0 \times 10^{-3}\ \text{N}$ to the right at P, what is the field at P?' Pause for calculation. Lead them through: $E = F / q_t$, show units: N / C . Compute: $6.0 \times 10^{-3}\ \text{N} \div 2.0 \times 10^{-6}\ \text{C} = 3.0 \times 10^3\ \text{N/C}$ to the right. Emphasize 'unit positive test charge' and 'without disturbing the source charges' by asking what would happen if the test charge were large (it would change the source charges — not allowed for the definition).

Board: Write the operational definition exactly as given. Then write the numeric worked example: 'Given: $F = 6.0 \times 10^{-3}\ \text{N}$ on $q_t = 2.0\ \mu\text{C}$ ($2.0 \times 10^{-6}\ \text{C}$). $E = F / q_t = 6.0 \times 10^{-3}\ \text{N} \div 2.0 \times 10^{-6}\ \text{C} = 3.0 \times 10^3\ \text{N/C}$ (direction: right).' Include units N/C .

- Anticipate: Why use a positive test charge? (A positive test charge gives the direction of the field in the usual sense; if you used a negative test charge the force would be opposite.)
- Anticipate: What does 'without disturbing the source charges' mean? (The test charge must be small enough that it does not change the arrangement of the source charges.)
- Anticipate: Is electric field a vector? (Yes — magnitude and direction matter; we showed direction 'to the right'.)

27-35 min: Introduce superposition for forces and link to field

Draw two source charges on the board: $+Q_1$ at left and $+Q_2$ at right, and a small positive test charge q_t at a point between them. Ask: 'What is the net force on q_t ?' Let students suggest 'add the forces from each charge.' Say: 'That is the principle of superposition: the net electrostatic force equals the vector sum of the forces from each source, and each pairwise force is unchanged by the others.' Work one simple numeric example: $Q_1 = +2e$ (so compute in coulombs), $Q_2 = +1e$, place $q_t = +1e$ at a point where force magnitudes from each are $4.8 \times 10^{-20}\ \text{N}$ left and $2.4 \times 10^{-20}\ \text{N}$ right respectively (use these numbers). Ask students to compute net force and direction: $\text{net} = 4.8 \times 10^{-20}\ \text{N}$ left minus $2.4 \times 10^{-20}\ \text{N}$ right = $2.4 \times 10^{-20}\ \text{N}$ left. Then show how to get net field by dividing net force by q_t : $E_{\text{net}} = F_{\text{net}} / q_t$ — here $q_t = 1e = 1.60 \times 10^{-19}\ \text{C}$ — compute E_{net} in N/C with units. Do the arithmetic aloud and write each step; keep units visible.

Board: Draw the diagram with $+Q_1$, $+Q_2$, and q_t . Write bullet: 'Superposition: $F_{\text{net}} = F_1 + F_2$ (vector sum).'

Then write the numeric steps showing q values in coulombs (convert using $n \cdot e$ in words), the given pairwise forces, the sum, and $E_{\text{net}} = F_{\text{net}} / q_t$ with the final number and units N/C .

- Anticipate: Why can we just add forces? (Because electrostatic forces follow the superposition principle; each pairwise force is calculated as if other charges were not present.)
- Anticipate: Do we need vectors? (Yes — direction matters; if forces are opposite we subtract magnitudes.)
- Anticipate: What if the charges are not pointlike? (Today's examples use point-like or small localized charges; distribution issues come later.)

35-38 min: Guided practice: compute charge and field

Give a quick two-part pair task (2 minutes): Part A: Convert $-5e$ to coulombs. Part B: A test charge of $+2e$ feels a force of $1.6 \times 10^{-19}\ \text{N}$ upward. What is the field at that point? Tell pairs: 'Write answers and one line of calculation; I'll call on two pairs.' Circulate, correct notation and units.

Call on one pair to present both answers. If they make arithmetic slip, write corrected steps on the board and ask class to spot the mistake.

Board: Write the two-part questions and leave space for student answers. After presentations, write the correct answers with calculation lines: ' $-5e = -8.0 \times 10^{-19} \text{ C}$ ' and ' $E = F/q = 1.6 \times 10^{-19} \text{ N} \div 3.2 \times 10^{-19} \text{ C} = 0.50 \text{ N/C}$ upward' (showing $q = 2e = 3.20 \times 10^{-19} \text{ C}$).

- Anticipate: How did you get $3.2 \times 10^{-19} \text{ C}$ for $2e$? (Multiply $2 \times 1.60 \times 10^{-19} \text{ C}$.)
- Anticipate: Why is the field smaller than the force? (Because field is force divided by the test charge; large test charge gives smaller E for same F .)
- Anticipate: Why is the sign negative for $-5e$? (Negative means an excess of electrons; the numerical value with sign gives direction for forces on positive test charges.)

38-40 min: Check for understanding and consolidate

Ask two quick checks aloud: 1) 'If a conductor is charged, where will excess charges end up?' Students answer: 'on the surface.' 2) 'If two charges both push a test charge, how do we find the total force?' Students: 'Add the vector forces (superposition).' Summarize: 'We used $q = n e$ to get coulombs, defined $E = F/q$ operationally, and used superposition to get net force or field. Next period we will treat source vs test charge more formally and compute flux.' End by asking students to keep today's board notes in their notebook.

Board: Underline the main points already on the board. Add one final bullet: 'Takeaway: q in C (use $n \cdot e$); $E = F/q$ (N/C); superposition = vector sum; conductors: surface; insulators: localized.'

- Anticipate: Will we need to memorize e exactly? (Yes — use $1.60 \times 10^{-19} \text{ C}$ here; more precision not needed for these problems.)
- Anticipate: How do we know when to divide by q to get E versus summing fields? (You can divide the net force by q or sum the fields from each source; both give the same E .)

Blackboard Notes

- Quantisation example
- Conductors vs Insulators
- Operational definition: Electric field
- Superposition and worked example
- Takeaway
- Example: $+3e \rightarrow 3 \times 1.60 \times 10^{-19} \text{ C} = 4.8 \times 10^{-19} \text{ C}$
- $e = 1.60 \times 10^{-19} \text{ C}$ (elementary charge magnitude)
- Conductor: charges move and redistribute $\rightarrow$ excess on surface
- Insulator: charges stay where placed (localized)
- Electric field (operational): force on a unit positive test charge placed at the point; equivalently force per unit test charge
- Units: force in newtons (N), charge in coulombs (C), field in N/C
- Superposition: net electrostatic force = vector sum of pairwise forces (each pairwise force unchanged by others)
- Takeaway: Use $n \cdot e$ to convert e -units to C ; $E = F / q$ to get field; always include units
- Diagram: Sketch: metal sphere with $+$ signs spread over outer surface; label 'conductor'
- Diagram: Sketch: plastic rod with $+$ signs clumped in one spot; label 'insulator'
- Diagram: Sketch: $+Q_1$ (left) and $+Q_2$ (right) with q_t between; arrows showing F_1 , F_2 on q_t and resulting F_{net}
- Formula: $F(\mathbf{r}) = q \mathbf{E}(\mathbf{r})$

Activities

Three Quick Problems: Conductors, Superposition, and Charge Quantisation (12 min, problem_set)

Materials: calculator, pencil and paper, ruler (cm/mm), textbook or reference with value of ϵ_0 (optional)

- 1. Arrange students in pairs. Say: "You have 12 minutes to complete three numbered problems; write every step and unit. I will circulate and check units and arithmetic."
- 2. Hand out or display the problems (student_instructions). Start the timer for 12 minutes.
- 3. While students work, circulate. For each pair, check that they (a) write the origin of Coulomb's law (proportionality then introduce $k = 1/(4\pi\epsilon_0)$), (b) show vector addition for forces, and (c) convert between n (elementary charges) and q (coulombs) with units.
- 4. With 3 minutes remaining, announce time left and ask pairs to box their final numeric answers with units and underline the step where they used superposition and the step where they converted charges using $q = n \cdot e$.
- 5. Collect papers or have pairs swap with another pair for quick peer check of units and arithmetic.

Student instructions:

- General: Write every algebraic step, carry units through each calculation, and give final numeric answers with units. Use $e = 1.60 \times 10^{-19} \text{ C}$ and $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$.
- Problem 1 — Conductor vs Insulator (qualitative, 3 min): You place a small metal sphere and a plastic sphere each carrying the same amount of excess negative charge. a) Describe where the excess electrons end up in the metal and draw a simple diagram showing charge distribution. b) Do the same for the plastic sphere (an insulator). c) State one experimental observation you would make that distinguishes the two (what you could see or measure).
- Problem 2 — Superposition of Forces on a Test Charge (quantitative, 5 min): Three point charges lie on the x-axis: $q_1 = +4.0 \text{ nC}$ at $x = 0.00 \text{ m}$, $q_2 = -2.0 \text{ nC}$ at $x = 0.30 \text{ m}$, and a small test charge $q_3 = +1.0 \text{ nC}$ at $x = 0.20 \text{ m}$. a) Convert every charge from nC to C showing the multiplication by e.g. $1 \text{ nC} = 1.0 \times 10^{-9} \text{ C}$. b) Write Coulomb's law beginning from the proportionality F proportional to $q_A q_B / r^2$, then introduce $k = 1/(4\pi\epsilon_0)$ and substitute the numeric value and units for k . c) Calculate the force on q_3 due to q_1 (vector, with sign and units). Show magnitude, direction, and units. d) Calculate the force on q_3 due to q_2 (vector). e) Use vector superposition to find the net force on q_3 ; show the vector sum step and the final numeric result with units.
- Problem 3 — Charge Quantisation and Order-of-Magnitude (quantitative, 4 min): A small object has a net charge of $q = -2.56 \times 10^{-17} \text{ C}$. a) Using $q = n \cdot e$, compute n (the integer number of excess electrons) and state whether this n must be an integer; show your division and units. b) Give an order-of-magnitude statement: approximately how many electrons is that? Write the answer in the form $n \sim ? \times 10^m$ electrons and show reasoning.
- End: Box each final numeric answer and underline where you used (i) superposition, and (ii) $q = n \cdot e$.

Success looks like: At least one student in each pair writes Coulomb's law starting from F proportional to $q_1 q_2 / r^2$ and then substitutes $k = 1/(4\pi\epsilon_0)$ with numeric value and units.; Each pair calculates both pairwise forces in Problem 2, shows their directions on the x-axis, and writes the vector sum step to produce a single net force with units.; Each pair converts charges to coulombs and computes n in Problem 3, showing the division q/e and giving an integer or nearest-integer statement.

Support: If stuck, use this sentence frame: “Coulomb’s law starts as F proportional to $q_1 q_2 / r^2$. Introducing k gives $F = k \cdot (q_1 q_2) / r^2$ where $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. For Problem 2, convert: $4.0 \text{ nC} = 4.0 \times 10^{-9} \text{ C}$. The distance r between a charge at $x=a$ and $x=b$ is $r = |b-a| \text{ m}$.” Example worked first step: For q_1 on q_3 , $r = |0.20 - 0.00| = 0.20 \text{ m}$; $q_1 = 4.0 \times 10^{-9} \text{ C}$, $q_3 = 1.0 \times 10^{-9} \text{ C}$; substitute into $F = k q_1 q_3 / r^2$ and compute units N. Extension: Harder: Replace q_2 in Problem 2 with a uniformly charged conducting rod of length 0.10 m centered at $x = 0.30 \text{ m}$ carrying total charge -2.0 nC . Derive an expression (with integral) for the x -component of the force on q_3 due to the rod: set up the integral, carry units through, and evaluate numerically. State any approximations used.

Checkpoint Questions

- [apply] A tiny object carries charge $q = +3.204 \times 10^{-19} \text{ C}$. Using $q = n e$ with $e = 1.602 \times 10^{-19} \text{ C}$, find n (the number of elementary charges) and state whether the object has excess protons or electrons.

Expected: Compute $n = q / e = (3.204 \times 10^{-19} \text{ C}) / (1.602 \times 10^{-19} \text{ C}) = 2$. So $n = +2$ (two positive elementary charges). The object has an excess of two protons (or is missing two electrons). Include units and the division step.

- [understand] You have a metal sphere and a plastic rod, each given the same net negative charge. Where does that extra negative charge reside on the metal sphere and on the plastic rod? Give one physical reason (in one sentence) for the difference.

Expected: On the metal sphere (a conductor) the extra negative charge redistributes and resides entirely on the surface. On the plastic rod (an insulator) the extra charge stays localized near where it was placed and does not move freely. Reason: in a conductor free electrons move until they reach the surface; in an insulator electrons are bound and cannot move freely.

- [apply] Two point charges lie on the x -axis: $+2e$ at $x = -0.10 \text{ m}$ and $-2e$ at $x = +0.10 \text{ m}$. Use superposition to find the direction and magnitude of the electric field at the origin ($x = 0$). (Use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ and $e = 1.602 \times 10^{-19} \text{ C}$.)

Expected: Field from $+2e$ at origin points to the right (away from $+$), magnitude $k(2e)/(0.10^2)$. Field from $-2e$ also points to the right (toward $-$), same magnitude. Net field $= 2 \times k(2e)/(0.10^2)$ to the right $= (4 k e)/(0.10^2)$. Numerically: $4 \times 8.99 \times 10^9 \times 1.602 \times 10^{-19} / 0.01 \sim 5.8 \times 10^{-7} \text{ N/C}$ to the right. Show vector direction and units.

EXIT TICKET

A small dust grain has measured charge $q = -4.806 \times 10^{-19} \text{ C}$. Using $q = n e$ with $e = 1.602 \times 10^{-19} \text{ C}$, compute n and explain in one sentence what that integer means physically.

Success indicator: Shows $n = q/e = (-4.806 \times 10^{-19} \text{ C}) / (1.602 \times 10^{-19} \text{ C}) = -3$, includes units and the arithmetic, and states that the grain carries a net charge of -3 elementary charges (three extra electrons).

Homework

- Convert each of these charges into an integer number of elementary charges (n) and show the arithmetic: (a) $q = +8.01 \times 10^{-19} \text{ C}$, (b) $q = -1.602 \times 10^{-18} \text{ C}$, (c) $q = +1.0 \times 10^{-21} \text{ C}$. State whether each result is physically allowed (integer) or indicates a measurement rounding.
- Two equal small metal balls are isolated and then touched together so charge can move. Before contact ball A had $+5e$ and ball B had $-1e$. After separating them, what is the charge on each ball? Show the conservation-of-charge calculation in elementary charges and in coulombs (use $e = 1.602 \times 10^{-19} \text{ C}$).

•

Short answer (one paragraph): Explain how you would use a small positive test charge to determine the direction of the electric field at a point near a charged insulating rod, mentioning what you must ensure so the test charge does not disturb the source charge distribution.

Estimated time: 30 min

MENTOR MOMENT — COUNTING CHARGES IN THE LAB

In a lab I once measured a tiny bead with $q \sim -1.602 \times 10^{-18} \text{ C}$ and first panicked; dividing by e showed it was exactly $-10e$. Realizing the charge was an integer made the result sensible and saved time troubleshooting the apparatus.

Takeaway: When a measured charge looks odd, divide by e and check whether you get a small integer — that often reveals measurement rounding or a correct quantised result.

Period 3

ENTRY TICKET

A plastic comb picks up small neutral bits of paper after you run it through your hair. Circle the best explanation and give one short sentence to justify it:

- A) The paper became net charged by touching the comb.
- B) The paper is still neutral but has its positive and negative charges moved slightly apart (induced dipole), and the comb attracts it.
- C) The paper produces its own electric field that pulls the comb in.

Write the letter and one sentence.

Expected response: B — The paper remains neutral overall but is polarised: charges separate so the nearer side is opposite in sign to the comb and attracted, giving a net attraction without net charge transfer.

Teacher Script

0-5 min: Collect entry tickets; brief class read-aloud

Say: "Take 2 minutes to finish your entry ticket and hand it to me when you're done." Walk the room quickly reading a few responses aloud: choose one that wrote A, one that wrote B. Then say: "If you wrote A, tell us why you thought the paper got a net charge." Pause for 1 quick student answer. Then say: "If you wrote B, say how the charges are arranged in the paper." Pause for 1 quick answer. Conclude: "Good — today we'll see why a neutral object can be attracted without gaining net charge: induced dipoles and how we describe fields and flux."

Board: Write the entry ticket question's key wording on the board and underline the correct choice B.

- Anticipate: But how can one side be more attracted if the paper has equal positive and negative charges?
- Anticipate: Does the comb give charge to the paper when we touch it?
- Anticipate: Is polarization permanent?

5-10 min: Operational definition of electric field and source/test charges

Say: "We need a clear definition: place a tiny positive test charge at a point and measure the force on it without moving the source charges. The electric field is that force per unit positive charge." Put a small toy ball labelled $+q_{\text{test}}$ and a larger fixed charged object on the desk as a physical anchor. Ask: "Which is the source charge and which is the test charge here?" Wait for answer. Then say: "The source charge must be treated as fixed; we only measure how a tiny test charge would feel a force there." Emphasize units: "Force in newtons divided by charge in coulombs gives E in newtons per coulomb (N/C)."

Board: Write the operational definition sentence exactly from the source and below it write units: $E = \text{N/C}$.

- Anticipate: How small must a test charge be?
- Anticipate: If the test charge moves, does it change the field?
- Anticipate: Why use a positive test charge?

10-15 min: Superposition principle (vector sum of fields)

Say: "If there are several point source charges, the net electric field at a point is the vector sum of the fields each source would produce there. We'll do a quick two-charge example." Draw two

point charges on the board: $+2 \times 10^{-6} \text{ C}$ at $x = -0.10 \text{ m}$ and $-1 \times 10^{-6} \text{ C}$ at $x = +0.10 \text{ m}$. Ask: "At the origin, what is the direction of the field from each charge?" Pause for answers: left / right. Then compute magnitudes step by step: show Coulomb's law verbally: 'force $\sim 1/r^2$ ' but do not write new formulas beyond allowed. Use numeric values: distance from each to origin is 0.10 m ; compute $E_1 = k \cdot 2 \times 10^{-6} / (0.10^2)$ and $E_2 = k \cdot 1 \times 10^{-6} / (0.10^2)$. Use $k = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$ and show units. Calculate numbers: $E_1 = (9.0 \times 10^9 \cdot 2 \times 10^{-6}) / (0.01) = (18 \times 10^3) / 0.01$? Wait compute: $9 \times 10^9 \cdot 2 \times 10^{-6} = 18 \times 10^3 = 1.8 \times 10^4$; divide by 0.01 gives $1.8 \times 10^6 \text{ N/C}$. Similarly $E_2 = 9 \times 10^9 \cdot 1 \times 10^{-6} / 0.01 = 9 \times 10^3 / 0.01 = 9 \times 10^5$? Check: $9 \times 10^9 \cdot 1 \times 10^{-6} = 9 \times 10^3 = 9000$; $/0.01 = 900000 = 9 \times 10^5 \text{ N/C}$. So net field is $E_{\text{net}} = E_1$ (to right because + charge repels positive test?) Wait sign careful: At origin, $+2 \times 10^{-6}$ at left (-0.1) pushes positive test to right. -1×10^{-6} at $+0.1$ attracts positive test to right as well. Both to right, so add magnitudes: $1.8 \times 10^6 + 9 \times 10^5 = 2.7 \times 10^6 \text{ N/C}$ to the right. Say each arithmetic step aloud and write short lines of the computation, keeping units. Ask: "What does superposition mean here?" Summarize: "Vector add component-wise if not collinear."

Board: Draw the two point charges with coordinates and arrows showing each field at the origin. Write the numerical calculation lines with $k = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$ and show $E_1 = 1.8 \times 10^6 \text{ N/C}$, $E_2 = 9.0 \times 10^5 \text{ N/C}$, $E_{\text{net}} = 2.7 \times 10^6 \text{ N/C}$ to the right.

- Anticipate: Why did both fields point the same way when one charge is negative?
- Anticipate: Where did the value $k = 9.0 \times 10^9$ come from?
- Anticipate: What if the point wasn't on the line between charges?

15-24 min: Introduce electric flux through a surface element

Say: "Flux tells us how many field lines pass through a surface. For a tiny surface element dS , flux $d\Phi = \mathbf{E} \cdot d\mathbf{S} = E dS \cos\theta$ where θ is angle between $\mathbf{E}$ and the area normal." Point to the board formula. Draw a small square surface element with a normal arrow and a field vector at an angle. Ask: "If $\mathbf{E}$ is parallel to the normal, what is $\cos\theta$ and what is the flux sign?" Wait for ' $\cos\theta = 1$, flux positive'. Ask: "If $\mathbf{E}$ is perpendicular to the normal?" Wait for 'flux zero.' Then do a quick numerical example: choose $E = 2.0 \times 10^4 \text{ N/C}$, area $dS = 2.0 \times 10^{-4} \text{ m}^2$, $\theta = 60^\circ$. Compute $d\Phi = E dS \cos 60^\circ = 2.0 \times 10^4 \cdot 2.0 \times 10^{-4} \cdot 0.5$. Do arithmetic aloud: $2.0 \times 10^4 \cdot 2.0 \times 10^{-4} = 4.0 \times 10^0 = 4.0$; times $0.5 = 2.0$. So $d\Phi = 2.0$ (units: $\text{N} \cdot \text{m}^2 / \text{C}$). Write units explicitly. Emphasize interpretation: "Positive flux means field leaves through the surface in the direction of the area normal."

Board: Write the flux formula exactly: $d\Phi = \mathbf{E} \cdot d\mathbf{S} = E dS \cos\theta$. Draw the surface element with normal, draw $\mathbf{E}$ at 60° , and show the numeric substitution and arithmetic leading to $d\Phi = 2.0 \text{ N} \cdot \text{m}^2 / \text{C}$.

- Anticipate: What are the units of flux?
- Anticipate: What does negative flux mean physically?
- Anticipate: How small must dS be for this formula to be valid?

24-32 min: Define electric dipole and quick calculation using quantisation $q = n e$

Say: "A dipole is two equal and opposite point charges $+q$ and $-q$ separated by $2a$. The dipole moment $\mathbf{p} = q \times 2a$ and points from negative to positive." Point to the board definition. Give a concrete example: let $q = 3 e$ (three elementary charges). Ask: "What is q in coulombs?" Wait for a student to say multiply by $e = 1.6 \times 10^{-19} \text{ C}$. Do it: $q = 3 \times 1.6 \times 10^{-19} \text{ C} = 4.8 \times 10^{-19} \text{ C}$. Next choose $2a = 2.0 \times 10^{-9} \text{ m}$ (separation). Compute magnitude $p = q \times 2a = 4.8 \times 10^{-19} \text{ C} \times 2.0 \times 10^{-9} \text{ m} = 9.6 \times 10^{-28} \text{ C} \cdot \text{m}$. Write units. Emphasize direction arrow from $-$ to $+$. Ask: "Why does a neutral object stuck to a charged comb not require net charge?" Link back to induced dipole: one side slightly more negative than the other so the comb attracts it. If time, demonstrate with small pieces of paper and a charged comb (safety permitting).

Board: Write the dipole definition sentence exactly and the formula $\mathbf{p} = q \times 2a$. Under it write the numerical steps:

$q = 3 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-19} \text{ C}$; $p = 4.8 \times 10^{-19} \times 2.0 \times 10^{-9} = 9.6 \times 10^{-28} \text{ C}\cdot\text{m}$. Draw a dipole diagram: $-q$ at left, $+q$ at right, arrow labelled p pointing right.

- Anticipate: Why point from negative to positive, not the other way?
- Anticipate: Is p a vector? What if charges are vertical?
- Anticipate: How does induced dipole strength depend on the material?

32-36 min: Guided practice and rapid consolidation

Hand out a 2-question worksheet (or display on the board): (1) A point $-1e$ charges at $x=0.05 \text{ m}$ and $+1e$ at $x=0.05 \text{ m}$; find q in C and p if separation $2a = 0.10 \text{ m}$. (2) A small surface element has area $1.0 \times 10^{-4} \text{ m}^2$, $E = 5.0 \times 10^3 \text{ N/C}$ perpendicular to the surface normal. Compute flux. Give students 4 minutes to work and circulate. After 4 minutes, call the class back and write the answers step-by-step on the board, asking students to tell the next arithmetic step. End with: "What is one thing you will remember from today?" Have 1-2 students answer briefly.

Board: Write the two worksheet questions on the board. After students work, write the solutions: $q = 1 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-19} \text{ C}$; $p = q \times 0.10 = 1.6 \times 10^{-20} \text{ C}\cdot\text{m}$. Flux = $E \, dS \cos 0 = 5.0 \times 10^3 \times 1.0 \times 10^{-4} = 0.50 \text{ N}\cdot\text{m}^2/\text{C}$.

- Anticipate: Should I use $e = 1.6 \times 10^{-19}$ or 1.602×10^{-19} ?
- Anticipate: Why did we pick the test charge positive in the definition?
- Anticipate: Can flux be larger than the field value times area?

36-40 min: Exit check and link to next lesson

Say: "Quick check: in one sentence, what is the dipole moment and what are its units?" Call on 2 students. Then say: "Next time we'll compare how dipole fields fall off with distance compared to single charges and begin Gauss's law applications. Tonight, read the first paragraph on dipole field fall-off if you can." Collect any remaining worksheets as students leave.

Board: Write two final prompts: 'Define dipole moment (one sentence)' and 'Units of flux and dipole moment'.

- Anticipate: Will we have homework on this?
- Anticipate: Will we calculate dipole fields in detail next time?

Blackboard Notes

- Operational definition of Electric Field
- Superposition example (two charges)
- Electric Flux through a surface element
- Electric Dipole and Dipole Moment
- Quick practice problems
- Electric field (operational): force on a unit positive test charge placed at the point (without disturbing the source charges); equivalently, force per unit test charge. Units: N/C .
- Superposition: $E_{\text{net}} = \text{vector sum of fields from each source charge evaluated at the point.}$
- Flux through element: $d\Phi = E \cdot dS = E \, dS \cos\theta$ ($\theta = \text{angle between } E \text{ and area normal}$). Units: $\text{N}\cdot\text{m}^2/\text{C}$.
- Dipole: two equal and opposite point charges $\pm q$ separated by $2a$. Dipole moment $p = q \, 2a$, directed from $-q$ to $+q$. Units: $\text{C}\cdot\text{m}$.
- Quantisation: $q = n \, e$. Use $e = 1.6 \times 10^{-19} \text{ C}$ for calculations.
- Diagram: Point source charge (label $+Q$) with a small positive test charge at a nearby point; arrow showing force on test charge.
- Diagram: Two point charges on x -axis: $+2.0 \times 10^{-6} \text{ C}$ at $x=-0.10 \text{ m}$ and $-1.0 \times 10^{-6} \text{ C}$ at

$x = +0.10$ m. Show field vectors at origin and arrows indicating directions.

- Diagram: Tiny surface element dS (square) with area normal arrow; show E making 60° with normal and the projection $E \cos\theta$ through the area.
- Diagram: Dipole: $-q$ at left, $+q$ at right, separation labelled $2a$; arrow from $-q$ to $+q$ labelled p .
- Formula:
$$\Phi = \mathbf{E} \cdot \mathbf{S} = E S \cos\theta$$
- Formula: $p = q \cdot 2a$

Activities

Hands-on: Two Charged Rods, One Test Paper — Superposition, Flux, Dipole (12 min, demonstration)

Materials: Two insulating rods or long plastic rulers (can be rubbed to charge) or one plastic rod and one glass/plastic ruler, Two different cloths for rubbing (e.g., wool and silk or fur and cloth), A handful of small pieces of tissue paper (about 1 cm squares) or small bits of aluminum foil, One metric ruler (30 cm), Masking tape or sticky tack, Meter stick or tape measure, Whiteboard and marker (or paper and pen) for short calculations, Stopwatch or phone timer

- Group students in pairs. Tell the class: “You will watch a 12-minute demonstration and then each pair will answer two short questions on the worksheet. The demonstration shows (1) source vs test charge, (2) superposition of fields from two charges, (3) electric flux through a small flat surface, and (4) what we call an electric dipole.”
- Prepare the two rods: rub one rod with one cloth and the other rod with the other cloth for 30 seconds each to give them opposite signs if materials are different (say plastic rubbed with wool becomes negative, glass with silk becomes positive). Place the rods on the table 40 cm apart, held by masking tape so they don't roll. While rubbing, say aloud which rod you expect to be positive or negative and why (anchor to rubbing materials).
- Demonstration part A — source and test charge (3 minutes): Hold one charged rod (source) stationary on the table. Pick up one small piece of tissue paper (test charge) with tweezers or fold it on a table and bring it slowly 5 cm from the rod. Ask a student pair to observe and report whether the tissue moves and in which direction. Repeat at 10 cm and 20 cm. While doing this, narrate: “The rod is the source charge; the paper is the test charge. We treat the rod as fixed so it defines the electric field.”
- Demonstration part B — superposition (3 minutes): Place one charged rod at $x = 0$ cm and the other at $x = 20$ cm (use the ruler to measure). Hold a single folded tissue paper at $x = 10$ cm (midpoint) at a fixed height and ask two different pairs to watch. Bring the rods in, keeping both fixed, and show whether the paper is attracted, repelled, or sees competing forces. Move the paper slightly left or right and ask students to describe the net motion. Verbally model adding vectors: point left force from left rod and right force from right rod and add them.
- Demonstration part C — flux and dipole geometry (4 minutes): Tape a small $2\text{ cm} \times 2\text{ cm}$ square of paper (area = $4\text{ cm}^2 = 4 \times 10^{-4}\text{ m}^2$) to the ruler so it stands as a flat area element at the midpoint between the rods, with its normal vector pointing along the line joining the rods. Ask students to imagine field lines crossing this element. First, show when only one rod is present (remove the other): ask pairs to predict direction of E and whether $E \cdot A$ is positive or negative using $E dA \cos\theta$ and state units ($\text{N} \cdot \text{m}^2/\text{C}$). Then restore both

rods (forming a dipole with charges $\pm q$ separated by $20\text{ cm} = 2a$). Explain the dipole moment vector direction (from negative to positive) and point it out physically. Finally, ask: is net flux through this small square zero, positive, or negative? Use the idea that $\text{flux} = E \cdot dS$ and that for a small element near the midpoint the incoming and outgoing lines from the two charges will usually not cancel perfectly — ask one pair to give the observation and one pair to write the quick estimate on the board.

- While the demonstration runs, circulate and ask each observed pair to state aloud: the sign of each rod, the direction of the test paper force, and whether the flux through the taped square is positive/negative/zero. After the demo, write on the board the quantisation relation $q = n e$ and do one short numeric conversion: if $q = 1.6 \times 10^{-6}\text{ C}$, how many elementary charges $n = q/e$? Show calculation and units on the board; ask a volunteer pair to perform the arithmetic (estimate acceptable).

Student instructions:

- You are in a pair. Watch the teacher's demonstration carefully and be ready to report three short observations to the teacher when asked.
- During Part A (source vs test): note which object is the source and which is the test charge. Record at each distance (5 cm, 10 cm, 20 cm) whether the tissue paper is attracted, repelled, or not noticeably affected.
- During Part B (superposition): at the midpoint between the two rods, say whether the paper feels a net leftward, net rightward, or nearly zero horizontal force. If there is movement when you nudge the paper left or right, describe which way it moves and why you think that shows vector addition of the two fields.
- During Part C (flux and dipole): look at the small taped square area element. State the direction of the area vector (normal) the teacher used and say whether the electric flux through that square is positive, negative, or near zero when (i) only one rod is present and (ii) both rods form a dipole. Write down the short numerical conversion the teacher shows (compute n for $q = 1.6 \times 10^{-6}\text{ C}$ using $e = 1.60 \times 10^{-19}\text{ C}$).

Success looks like: Each pair can identify which object in Part A was the source and which the test charge and can point to the distance where the test paper showed the strongest attraction.; At least one student in each pair can indicate the two individual force directions during Part B (arrows left and right) and can state the observed net direction at the midpoint.; Each pair can state whether the small taped square's flux sign (positive/negative/near zero) changed when going from one rod to both rods and can point to the area vector direction used in the demonstration.

Support: Pair a student who struggles with a student who explains observations. Give this sentence frame: "The source is ____ because _____. The test charge moved ____ at distance ____ cm, so the field there is toward/away from the source." Work the conversion example together: $q = 1.6 \times 10^{-6}\text{ C}$; $e = 1.60 \times 10^{-19}\text{ C}$; $n = q/e = (1.6 \times 10^{-6}) / (1.60 \times 10^{-19}) = 1.0 \times 10^{13}$ elementary charges.

Extension: For an early-finisher: use the ruler measurement $2a = 20\text{ cm}$ and assume each rod carried charge magnitude $q = 1.6 \times 10^{-6}\text{ C}$ (use that number). Estimate the electric field magnitude at a point on the line perpendicular to the midpoint at a distance $r = 50\text{ cm}$ from the midpoint (treat the pair as a dipole). Show the step-by-step vector sum of fields from each charge (include units N/C), then compute an approximate dipole moment $p = q(2a)$ in coulomb-meter and express p in SI units.

Checkpoint Questions

- [apply] A $+4.8 \times 10^{-19}$ C point charge sits at the origin and a -4.8×10^{-19} C point charge sits at $x = 2.0$ cm. Using the operational definition of E , what is the direction and sense (toward/away) of the net electric field at the point $x = 1.0$ cm on the x -axis? Show reasoning in one line.

Expected: Both charges produce fields on the x -axis. At $x = 1.0$ cm the field from the $+$ charge (at 0) points away from origin (toward $+x$) and the field from the $-$ charge (at 2.0 cm) points toward the negative charge (also toward $+x$). They add in the $+x$ direction, so the net E points $+x$. (Student should note both contributions point same way and add.)

- [apply] Write the expression for the electric flux through a flat square of area 0.020 m^2 when a uniform field of magnitude 200 N/C passes through it at 60° from the area normal. Calculate the numerical flux with units.

Expected: $d\Phi = E dS \cos\theta$. Numerically: $\Phi = (200 \text{ N/C})(0.020 \text{ m}^2)\cos 60^\circ = 200 \times 0.020 \times 0.5 = 2.0 \text{ N}\cdot\text{m}^2/\text{C}$.
(Include units $\text{N}\cdot\text{m}^2/\text{C}$.)

- [apply] A dipole consists of $+2.0e$ and $-2.0e$ separated by 1.0 cm (so $2a = 1.0$ cm). Give the dipole moment magnitude in coulomb-meter and state its direction using the definition in the lesson.

Expected: $p = q \cdot 2a = (2.0 \times 1.602 \times 10^{-19} \text{ C}) \times (0.010 \text{ m}) = 3.204 \times 10^{-21} \text{ C}\cdot\text{m}$ ($\sim 3.2 \times 10^{-21} \text{ C}\cdot\text{m}$). Direction: from the negative charge to the positive charge.

EXIT TICKET

An object carries a net charge of -9.6×10^{-19} C. Using $q = n e$, (a) how many excess elementary charges n does the object have (give sign and integer or nearest integer), and (b) express that charge as a multiple of e (e.g., $-3 e$). Show the short calculation.

Success indicator: Student divides charge by elementary charge $e = 1.602 \times 10^{-19} \text{ C}$ to get $n \sim -6.00$ (so $n = -6$ excess electrons) and writes the charge as $-6 e$ or $-9.6 \times 10^{-19} \text{ C} = (-6)(1.602 \times 10^{-19} \text{ C})$.

Homework

- 1) Convert the following numbers of elementary charges to coulombs (give answers in C): (a) $+1.0 \times 10^6 e$, (b) $-3 e$, (c) $+12 e$. Show the division by $e = 1.602 \times 10^{-19} \text{ C}$.
- 2) Three point charges lie on the x -axis: $+2.0 \times 10^{-19} \text{ C}$ at $x=0$, $-1.0 \times 10^{-19} \text{ C}$ at $x=3.0$ cm, $+1.0 \times 10^{-19} \text{ C}$ at $x=6.0$ cm. Compute the net electric field (magnitude and direction) at $x=9.0$ cm by vector superposition; show intermediate values and units (N/C).
- 3) A uniform electric field of 150 N/C is perpendicular to a circular disk of radius 4.0 cm. (a) Calculate the flux through the disk. (b) If the same field is tilted so the angle between E and the area normal is 30° , what is the new flux? Show formulas and units.

Estimated time: 40 min

MENTOR MOMENT — COMB, PAPER AND INDUCED DIPOLES

I once rubbed a plastic comb and watched tiny bits of paper jump up. The comb didn't give the paper a net charge; the comb's field pulled the paper's charges slightly apart (an induced dipole), and the nearer opposite charges felt a stronger pull, so the paper stuck.

Takeaway: A neutral object can be attracted without net charging: induced dipoles in a non-uniform field produce a net force toward the charged object.

Period 4

ENTRY TICKET

Look at a pair of equal and opposite point charges $+q$ and $-q$ separated by $2a$ on the x-axis, centered at the origin. At a point far away on the $+x$ axis (distance r from origin, with $r \gg a$), do you expect the electric field magnitude to fall off like $1/r^2$ or $1/r^3$? Circle one and write one short sentence giving your reason.

Expected response: Circle $1/r^3$. Reason (short): At large r the $+q$ and $-q$ nearly cancel their $1/r^2$ fields, leaving a weaker field that falls off like $1/r^3$; the field depends on the dipole moment $p = q(2a)$.

Teacher Script

0-5 min: Collect tickets & quick share

Say: "Put your paper face-up on the desk so I can see your answer. I'll call on a few students. If you circled $1/r^2$, be ready to say one sentence why; if you circled $1/r^3$, be ready to explain what 'dipole moment' might mean." Walk to three students, read one answer aloud and ask: "Which of you wrote the opposite? Tell us your one-sentence reason." Keep the tone neutral. If a student says 'they cancel so field is zero', ask: "Is it zero for every r , or only in a special place?"

Board: Write on board header: Entry ticket answers: $1/r^2$ or $1/r^3$? Note common answers.

- Anticipate: Why would the fields cancel? Aren't they different signs so they always cancel?
- Anticipate: If charges are equal and opposite, shouldn't the field always be zero?
- Anticipate: What exactly counts as 'far away' (how big must r be compared to a)?

5-15 min: Derive dipole far-field scaling ($r \gg a$)

Say: "We'll derive how the field behaves far from a dipole made of $+q$ at $x=+a$ and $-q$ at $x=-a$. Keep units: q in coulombs, distances in meters." Draw the two charges on the board. Step 1: Write Coulomb's law and point to each charge: use F formula as force and then convert to E by dividing by a test charge. Say: "Electric field of a point charge at distance R is $E = (1/4\pi\epsilon_0) q / R^2$ in direction $\hat{r}$." Step 2: For a point on the $+x$ axis at coordinate $x = r$ ($r \gg a$), write distances to each charge: $R_+ = r - a$, $R_- = r + a$. Step 3: Write the fields from each: $E_+ = (1/4\pi\epsilon_0) q / (r - a)^2$ to the right; $E_- = (1/4\pi\epsilon_0) (-q) / (r + a)^2$ to the right (explain sign). Step 4: Add $E = E_+ + E_-$; show algebraic subtraction: factor $(1/4\pi\epsilon_0) q [1/(r - a)^2 - 1/(r + a)^2]$. Step 5: Expand to leading order in a/r by using binomial approximation: $1/(r \pm a)^2 \sim 1/r^2 (1 \mp 2a/r + 3(a/r)^2 \dots)$. Keep only first-order term in a/r after subtraction. Show result $E \sim (1/4\pi\epsilon_0) * (q * 4a) / r^3$. Then identify $p = q(2a)$ and write E proportional to $p / (4\pi\epsilon_0 r^3)$. Emphasize units: p in $C \cdot m$, E in N/C . Conclude: falls off as $1/r^3$ for $r \gg a$.

Board: Write each step equations exactly as you derive: distances $R_{\pm} = r \mp a$; $E = (1/4\pi\epsilon_0) q [1/(r - a)^2 - 1/(r + a)^2]$; show binomial expansion steps; final form proportional to $p / (4\pi\epsilon_0 r^3)$.

- Anticipate: Why can we drop the higher order terms in a/r ? How big is a/r allowed to be?
- Anticipate: Where did the $4a$ come from in the numerator?
- Anticipate: Why do we define $p = q(2a)$ and not $q a$?

15-22 min: Short demo: measure idea of faster falloff

Say: "I'll show a quick demonstration with two small charged pith balls fixed at $\pm a$ and a small test probe on a rod. You will watch how the probe reading changes when I move it away." Place

two opposite charges separated by known $2a$ (say $a = 1\text{ cm}$ so $2a = 2\text{ cm}$) and bring a small calibrated field sensor (or an electroscope with marked distances) along the axis; announce distances $r = 5\text{ cm}, 10\text{ cm}, 20\text{ cm}$. At each r , read sensor and say expected scaling: if $1/r^2$ expected, ratio from $5 \rightarrow 10\text{ cm}$ would be $1/4$; if $1/r^3$ expected ratio is $1/8$. Ask students to watch which ratio matches experiment more closely. After readings, state observation: readings drop faster than $1/r^2$ consistent with $1/r^3$ scaling for $r \gg a$.

Board: Write the measured sensor values and the ratios between distances; note comparison to $1/r^2$ and $1/r^3$ expected ratios.

- Anticipate: Our sensor readings are noisy — how do you know it's $1/r^3$ and not experimental error?
- Anticipate: What if I measure off the axis, does it still fall like $1/r^3$?
- Anticipate: Can we do this with bigger charges so it's clearer?

22-32 min: Gauss's law quick review and line-charge example

Say: "Quick Gauss's law reminder: the flux through a closed surface equals enclosed charge divided by ϵ_0 . Write the formula and then use a known example: infinite line charge." Point to the cylindrical Gaussian surface diagram. Ask: "What symmetry simplifies E on the curved side?" (Answer: radial and same magnitude). Walk through applying Gauss: flux through curved surface $= E(2\pi r L)$ and flux through end caps $= 0$; enclosed $q = \lambda L$. Solve for $E = \lambda/(2\pi\epsilon_0 r)$. State units: λ in C/m , r in m , E in N/C . Emphasize this is a different falloff ($1/r$) from dipole and monopole cases and explain why symmetry and infinite extent change result.

Board: Write Gauss's law formula; draw cylinder with line charge; write flux $= E(2\pi r L) = q_{\text{enc}}/\epsilon_0$ and solve $E = \lambda/(2\pi\epsilon_0 r)$.

- Anticipate: Why are the end caps flux zero? Don't field lines hit them?
- Anticipate: What does 'infinite line' mean practically — how long must it be?
- Anticipate: Is Gauss's law always true even if field is not symmetric?

32-37 min: Guided practice: numerical charge calculation $q = n e$

Say: "Now a short calculation: If a dust particle has net charge $+20e$, what is q in coulombs? Use $e = 1.60 \times 10^{-19}\text{ C}$. Work with your partner and write the answer with units and an order-of-magnitude statement." Circulate for 2.5 minutes. After partners finish, call a student to write result: $q = 20 \times 1.60 \times 10^{-19}\text{ C} = 3.2 \times 10^{-18}\text{ C}$. Ask: "What is an order-of-magnitude estimate?" (Answer: about 10^{-18} C). Emphasize carrying units through.

Board: Write the worked conversion: $q = n e = 20 \times 1.60 \times 10^{-19}\text{ C} = 3.2 \times 10^{-18}\text{ C}$. Note: $e = 1.60 \times 10^{-19}\text{ C}$.

- Anticipate: Why do we multiply by e ? Is charge always a multiple of e ?
- Anticipate: What about electrons (negative sign)? Should we put a minus?
- Anticipate: How to estimate if n is very large like 10^9 ?

37-40 min: Quick check and consolidation

Say: "Two quick questions on your paper: (1) For a dipole with $p = q(2a) = 4.0 \times 10^{-9}\text{ C}\cdot\text{m}$, at $r = 0.10\text{ m}$ along the axis, what is the expected scaling of the field with r (write only ' $1/r^2$ ' or ' $1/r^3$ ')? (2) Convert $-5e$ to coulombs. Write answers and hand up." Wait 60–90 seconds, then read correct answers: (1) $1/r^3$ (2) $-8.0 \times 10^{-19}\text{ C}$. Finish: "Today we used Coulomb's law to get dipole far-field $\sim p/(4\pi\epsilon_0 r^3)$ and reviewed Gauss's law example for a line charge. Keep these results for next time when we draw field lines and compute torques."

Board: Write final two quick-check questions and correct answers briefly.

- Anticipate: For the quick check, why didn't you ask for the constant factor in the dipole

field?

- Anticipate: Will we need the exact prefactor $p/(4\pi\epsilon_0 r^3)$ later or just the $1/r^3$ dependence?
- Anticipate: Can we use Gauss's law to get the dipole field exactly?

Blackboard Notes

- Dipole far-field ($r \gg a$)
- Coulomb's law
- Gauss's law and infinite line-charge
- Charge quantisation
- Dipole: two opposite charges $+q$ and $-q$ separated by $2a$, center at origin.
- For point on axis at $x = r$ with $r \gg a$, distances: $R_+ = r - a$, $R_- = r + a$ (units: m).
- Add point-charge fields and expand in a/r to leading order: E proportional to $p/(4\pi\epsilon_0 r^3)$ with $p = q(2a)$ (p in $\text{C}\cdot\text{m}$).
- Dipole field falls off as $1/r^3$ for $r \gg a$ (not $1/r^2$).
- Coulomb's law (point charges): $F = (1/4\pi\epsilon_0) (q_1 q_2 / r^2) \hat{r}$ (N, C, m).
- Gauss's law (integral): $\Phi_E = \text{closed integral } \mathbf{E} \cdot d\mathbf{S} = q_{\text{enc}} / \epsilon_0$.
- Infinite line charge example: choose cylinder coaxial with line. Flux through curved surface $= E(2\pi r L)$. Enclosed $q = \lambda L$.
- Solve: $E = \lambda/(2\pi\epsilon_0 r)$ (units: λ in C/m , E in N/C).
- Charge quantisation: $q = n e$, with $e = 1.60 \times 10^{-19} \text{ C}$. Example: $20e = 3.20 \times 10^{-18} \text{ C}$.
- Diagram: Diagram: x-axis with $-q$ at $x=-a$ and $+q$ at $x=+a$; point at $x=r$ labelled r ($r \gg a$).
- Diagram: Diagram: cylindrical Gaussian surface around an infinite line-charge (show radius r and length L).
- Diagram: Small table of sensor readings vs distance for demo ($r = 5 \text{ cm}, 10 \text{ cm}, 20 \text{ cm}$) with ratios.
- Formula: $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$
- Formula: $\Phi_E = \oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{q_{\text{enc}}}{\epsilon_0}$

Activities

Desk Simulation: Compare Monopole and Dipole Fields, Flux, and Charge Quantisation (10 min, simulation)

Materials: worksheet printed one per pair (see steps below), scientific calculator per student, ruler or metre stick, stopwatch or timer visible to class, whiteboard and marker

1. Arrange students in pairs. Hand each pair the printed worksheet that lists the numerical scenario: a point charge $+q$ and a point charge $-q$ separated by $2a = 2.0 \text{ cm}$ (so $a = 1.0 \text{ cm}$), with $q = 5e$ (five elementary charges). Tell students they have 10 minutes and start the timer. Say: "We will compute fields at three distances, compare fall-off rates, check flux through a sphere, and convert charge units."
2. Tell pairs to follow the worksheet step sequence. While they work, circulate and check units are carried (e.g. distances in metres, charge in coulombs). If a pair finishes early, give them the extension question on the whiteboard. If a pair is stuck on starting values, read the worked example in the Support aloud once.
3. After 8 minutes announce: "Prepare one clear sentence stating whether the dipole field falls faster than the monopole and give one numerical ratio you found." Call on two pairs to

read their sentence and ratio in 30 seconds each. Record one pair's computed flux value on the board.

- 4. At the end of 10 minutes collect the worksheets and note pairs that used correct unit conversions and who produced the numerical comparisons. Do not lecture; use student answers to plan next class.

Student instructions:

- You are a pair. Use the calculator and show units at every step. Convert centimetres to metres when required ($1 \text{ cm} = 0.01 \text{ m}$). Use the elementary charge $e = 1.60 \times 10^{-19} \text{ C}$ and $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ (or $1/(4\pi\epsilon_0)$).
- Step A — Charge conversion (2 minutes): Compute q in coulombs using $q = n e$ with $n = 5$. Show q in C. (Example: $q = 5 \times 1.60 \times 10^{-19} \text{ C} = ? \text{ C}$).
- Step B — Monopole comparison (3 minutes): Treat a single $+q$ at the origin. Compute the electric field magnitude E_{mono} at $r_1 = 0.05 \text{ m}$, $r_2 = 0.15 \text{ m}$, and $r_3 = 0.50 \text{ m}$ using Coulomb's law $E = k|q|/r^2$. Write each E with units N/C and record ratios $E(r_1)/E(r_2)$ and $E(r_2)/E(r_3)$.
- Step C — Dipole approximation (3 minutes): Model a dipole formed by $+q$ at $x = +a$ and $-q$ at $x = -a$ with $a = 0.01 \text{ m}$. For points on the x -axis at distances $R = 0.05 \text{ m}$, 0.15 m , and 0.50 m from the origin ($R \gg a$), compute the approximate dipole field magnitude using the leading-order far-field expression derived from Coulomb's law for two charges: $E_{\text{dip}} \sim (1/(4\pi\epsilon_0)) \cdot (2qa)/R^3$. Derive this form briefly by adding the fields from $+q$ and $-q$ along the axis and keeping the first nonzero term in a/R . Use $p = q(2a)$ and write E_{dip} in N/C. Record the ratios $E_{\text{dip}}(R_1)/E_{\text{dip}}(R_2)$ and $E_{\text{dip}}(R_2)/E_{\text{dip}}(R_3)$.
- Step D — Compare fall-off (1 minute): Compare the numerical ratios from Steps B and C and state which falls off faster with R ($1/R^2$ vs $1/R^3$). Give one numerical example (a ratio) to support your statement.
- Step E — Gauss's law check (1 minute): Imagine a spherical surface of radius 0.02 m centered at the origin. Decide whether the total enclosed charge is zero or nonzero for (i) the monopole case where only $+q$ at origin exists, and (ii) the dipole case with $+q$ and $-q$ at $\pm a$. Use $\Phi = Q_{\text{enclosed}}/\epsilon_0$ to compute the flux Φ in each case and give units ($\text{N}\cdot\text{m}^2/\text{C}$).

Success looks like: Each pair shows q in coulombs with correct multiplication by e and correct units (e.g. $q = 8.00 \times 10^{-19} \text{ C}$); Each pair writes three monopole E values with units N/C and three dipole E_{dip} values with units N/C, and can point to the ratio demonstrating a $1/R^2$ versus $1/R^3$ trend (monopole ratios $\sim (R_2/R_1)^2$, dipole ratios $\sim (R_2/R_1)^3$); Each pair correctly states the enclosed charge and writes the numerical flux Φ with units $\text{N}\cdot\text{m}^2/\text{C}$ for the sphere examples.

Support: If you are unsure how to start, use this worked first item: $q = 5e = 5 \times 1.60 \times 10^{-19} \text{ C} = 8.00 \times 10^{-19} \text{ C}$. Monopole at $r = 0.05 \text{ m}$: $E = k|q|/r^2 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(8.00 \times 10^{-19} \text{ C})/(0.05 \text{ m})^2 =$ calculate result and write N/C. Sentence frame for comparisons: "At $R = ___ \text{ m}$, $E_{\text{mono}} = ___ \text{ N/C}$ and $E_{\text{dip}} = ___ \text{ N/C}$. The ratio $E_{\text{mono}}(R_1)/E_{\text{mono}}(R_2) = ___ \sim (R_2/R_1)^2$, while $E_{\text{dip}}(R_1)/E_{\text{dip}}(R_2) = ___ \sim (R_2/R_1)^3$." Pair lower-step student with one stronger peer if stuck.

Extension: Change q to $1.0 \times 10^{-9} \text{ C}$ and increase a to 0.10 m . Compute the dipole field at $R = 5.0 \text{ m}$ using both the exact numeric addition of two point-charge fields (sum fields from $+q$ at $+a$ and $-q$ at $-a$) and the dipole approximation $E_{\text{dip}} \sim (1/(4\pi\epsilon_0)) \cdot (2qa)/R^3$. Report both values, compute the percentage difference, and explain why the approximation improves or worsens when R changes relative to a . Show all units and steps.

Checkpoint Questions

- [apply] A particle carries charge $q = -3.2 \times 10^{-19} \text{ C}$. Using $q = n e$ where $e = 1.60 \times 10^{-19} \text{ C}$, find n (the number of elementary charges) and say whether this q could come from an integer number of electrons.

Expected: $n = q/e = (-3.2 \times 10^{-19} \text{ C}) / (1.60 \times 10^{-19} \text{ C}) = -2$. So $n = -2$, which is an integer: the charge equals the charge of two electrons, so it is physically possible.

- [apply] At points far from a dipole (distance r much larger than dipole half-separation a), the dipole electric field scales roughly as $1/r^3$. Suppose at $r = 1.0 \text{ m}$ the field magnitude is $E_1 = 2.0 \times 10^{-6} \text{ N/C}$. Estimate the field at $r = 2.0 \text{ m}$ and show the reasoning with units.

Expected: Dipole field scales as $1/r^3$ so $E_2 = E_1 \times (r_1/r_2)^3 = 2.0 \times 10^{-6} \text{ N/C} \times (1.0/2.0)^3 = 2.0 \times 10^{-6} \text{ N/C} \times 1/8 = 2.5 \times 10^{-7} \text{ N/C}$. Units remain N/C.

- [understand] You choose a cylindrical Gaussian surface coaxial with an infinite line charge of linear charge density λ . Write the Gauss's law equation that relates the magnitude E on the curved surface to the enclosed charge and solve algebraically for $E(r)$ (show units for λ and E).

Expected: Flux $\Phi = E \times (\text{curved area}) = E \times (2\pi r L)$. Enclosed charge $q_{\text{enc}} = \lambda L$ (λ in C/m). Gauss's law: $E(2\pi r L) = q_{\text{enc}}/\epsilon_0 = \lambda L/\epsilon_0$. Cancel L : $E(r) = \lambda/(2\pi \epsilon_0 r)$. Units: λ [C/m], ϵ_0 [$\text{C}^2/(\text{N}\cdot\text{m}^2)$] so E [N/C].

EXIT TICKET

A small bead has total charge $q = +4.8 \times 10^{-19} \text{ C}$. (a) Express this charge as n times the elementary charge $e = 1.60 \times 10^{-19} \text{ C}$ and give n with sign. (b) State in one sentence whether this q could come from an integer number of protons or electrons and why.

Success indicator: Shows calculation $n = q/e = (4.8 \times 10^{-19}) / (1.60 \times 10^{-19}) = +3$ and states that yes: $n = +3$ is an integer so the charge could come from three excess protons (or a deficit of three electrons).

Homework

- Convert each listed charge into the integer number of elementary charges n and into coulombs if needed: (a) $+1.60 \times 10^{-19} \text{ C}$, (b) $-4.8 \times 10^{-19} \text{ C}$, (c) $+8.0 \times 10^{-21} \text{ C}$. Show the division q/e with units and state whether each is physically an integer multiple of e .
- Estimate the number of electrons that make up a net charge of $-1.0 \mu\text{C}$ (microcoulomb). Show the calculation and give the answer in scientific notation.
- A dipole has charges $\pm q$ separated by $2a$. If at $r = 0.50 \text{ m}$ the dipole field magnitude is $8.0 \times 10^{-6} \text{ N/C}$, estimate the field at $r = 1.50 \text{ m}$ using the $1/r^3$ scaling and show units.

Estimated time: 25 min

MENTOR MOMENT — SMALL NUMBERS, BIG MEANING

In a lab, a student expected tiny charges like 10^{-19} C to be measurement noise, then found their value was exactly $-2e$ after division by e and recognized two missing electrons from a transfer. That integer result made the tiny number meaningful.

Takeaway: Whenever you get a tiny charge, divide by e and check for an integer — an exact integer tells you how many elementary charges were transferred.

Period 5

ENTRY TICKET

On the desk paper, draw quickly (1) a positive point charge and (2) a neutral small piece of paper near a charged comb. For each drawing, next to it write: 'Field direction near the charge?' and 'Does the paper become net charged or just pulled?'. Spend 3 minutes.

Expected response: Students draw radial arrows away from the + charge. For the paper+comb, many will write 'paper is attracted and becomes charged' or 'paper gets pulled (maybe charged)'. Correct responses to look for: radial field arrows for the + charge; for the paper, the correct explanation is 'paper is polarized (induced dipole) and is attracted without gaining net charge.'

Teacher Script

0-5 min: Collect quick ideas; correct visible misconceptions

Look at three students' quick drawings. Say: "Thanks — two quick checks. First drawing: point charge. Who drew field arrows? Hold up your paper. These arrows point away from a positive charge — good. Second: paper and comb — show one. I see answers saying 'paper becomes charged' and others saying 'paper is pulled.' Both are common. We'll clarify this now: neutral objects can be attracted without gaining net charge because the external field shifts charges inside the object and creates an induced dipole. I'll show a short demo in a moment." Pause to let students nod. Ask: "Why are radial arrows drawn? What do arrows show — direction or strength?" If students say direction, affirm; if someone says number of arrows, ask "Would drawing twice as many arrows change the real field?" and wait for responses to surface the count misconception.

Board: Write two tiny student sketches pinned to the board: (1) a + with radial arrows; (2) a comb near paper with a note 'answers varied'. Underline the words 'induced dipole' and 'field direction vs. drawn lines'.

- Anticipate: Student: So the paper never gets any extra charge? — Answer: It stays net neutral unless charge is actually transferred. The field separates charges inside it.
- Anticipate: Student: But why do diagrams sometimes show many lines? — Answer: The number drawn is a choice; we use density of lines per area to indicate relative strength, not the absolute count.

5-17 min: Introduce electric field and point-charge formula

Say: "We define the electric field E as a vector field produced by a charge that tells us the force per unit test charge at each point. Physically, if a small test charge q_0 is placed at r , the force is $F = q_0 E(r)$." Point to a coordinate origin on the board. Continue: "For a point charge Q at the origin the field at position r is this vector (I will write it and then explain each part)." Write the formula shown below. Then read aloud each symbol: " $1/(4\pi\epsilon_0)$ is a constant with units $N\cdot m^2/C^2$; Q is charge in coulombs C ; $\hat{r}$ is the unit vector pointing from the charge to the field point; r is the distance in meters. This gives E in units N/C (which is the same as V/m)." Ask: "If Q is positive, which way does $\hat{r}$ point and which way does E point?" Wait for answer (away). Then do a quick numeric anchor: "If $Q = +2.0\ \mu C$ and a point is $0.50\ m$ away, we can compute the magnitude: plug numbers: $k = 8.99 \times 10^9\ N\cdot m^2/C^2$, $Q = 2.0 \times 10^{-6}\ C$, $r = 0.50\ m$. Calculate magnitude on the board and show units. (Do the arithmetic step-by-step.)"

Board: Write the point-charge formula (see formulae). Below it, write the numeric example: $k = 8.99e9$

$\text{N}\cdot\text{m}^2/\text{C}^2$; $Q = 2.0\text{e-}6 \text{ C}$; $r = 0.50 \text{ m}$; compute $E = k Q / r^2$ with units, and the numerical result.

- Anticipate: Student: Why N/C equals V/m ? — Answer: Give quick statement: 1 N/C is the force per unit charge, and a volt per meter is the potential change per distance; they are equivalent units for E here.
- Anticipate: Student: Why use $\hat{r}$? — Answer: To emphasize E is a vector; $\hat{r}$ gives direction (away from +, toward -).

17-25 min: Field lines, density, and flux with small-area element

Say: "Now field lines are a helpful picture. Remember: a field line's tangent gives the local field direction. Density of lines per area perpendicular to the field shows relative strength — not the raw count of lines we draw." Draw on board: a positive charge with a few lines; then show the same with twice the drawn lines and explain both are valid; only density matters. Next: introduce flux. Draw a small flat patch with normal $\hat{n}$ and label its area dS . Say: "Flux through this element is $d\Phi = E \cdot dS$. Here dS is the vector $\hat{n}$ times the area magnitude (units m^2). If E is perpendicular and uniform over the patch, $d\Phi = E dS$. If at an angle, it's $E dS \cos \theta$." Ask: "Why would flux be useful?" Accept answers like 'counts how much field passes through surface' and link to Gauss's law students saw before.

Board: Draw: two versions of field-line diagrams around + (few lines and many lines) with an arrow 'density, not absolute count'. Draw a small rectangular patch, show $\hat{n}$, label dS , write $d\Phi = E \cdot dS$ and note $dS = \hat{n} dS$ and units.

- Anticipate: Student: If I draw more lines will the flux change? — Answer: No — drawing more lines is only a drawing choice; flux is calculated from E and dS values.
- Anticipate: Student: What if the field varies across the patch? — Answer: Then break into smaller patches and integrate (we'll do integrals next periods).

25-35 min: Demo: charged rod, paper bits, and induced dipole; torque on a dipole

Perform demo: bring a charged PVC rod or comb and small bits of paper. Show attraction of paper bits. Say: "Watch orientation of bits as they approach: they align such that nearest charges attract. This shows induced polarization, not net charging." Next, draw a simple dipole: $+q$ at $x = +a$ and $-q$ at $x = -a$. Label dipole moment p pointing from negative to positive. Say: "In a uniform field E a dipole feels a torque that tries to align p with E , but no net force because forces on $+q$ and $-q$ cancel. The torque is $\tau = p \times E$. I'll write that formula exactly as given." Write the torque formula and explain cross product direction using right-hand rule physically: "If p is to the right and E upward, torque points out of the board — that means it will rotate the dipole." Then ask: "Would a dipole in a non-uniform field have a net force?" Guide: "Yes, because the magnitudes on the two charges differ with position, producing a net force."

Board: Draw charged rod and paper bits photo/sketch; draw dipole $+q$ and $-q$ on x -axis and arrow p from $-q$ to $+q$; write $\tau = p \times E$ and note 'uniform $E \rightarrow$ torque only, no net force; non-uniform $E \rightarrow$ possible net force'.

- Anticipate: Student: Why doesn't the dipole get pushed along in a uniform field? — Answer: Because the forces on $+q$ and $-q$ are equal and opposite in a uniform field; they make a torque but sum to zero.
- Anticipate: Student: How big is p ? — Answer: $p = q \cdot \text{separation vector}$ (we won't derive units now, but p has units $\text{C}\cdot\text{m}$).

35-40 min: Guided practice and check for goals

Hand out a short worksheet (or project it) with two quick problems: (A) Three point charges: $q_1 = +2.0 \mu\text{C}$ at $x=0$, $q_2 = -1.0 \mu\text{C}$ at $x=0.30 \text{ m}$. Find E at $x = 0.15 \text{ m}$ (midpoint) and state direction. (B) A

dipole with $p = 6.0 \times 10^{-9} \text{ C}\cdot\text{m}$ in a uniform $E = 5.0 \times 10^3 \text{ N/C}$ upward. Find torque magnitude and will it translate? Give 5 minutes for students to work in pairs. Circulate and prompt: "Remember vector addition for fields, and use $\tau = pE \sin\theta$ for magnitude". After 4 minutes, collect one answer from the class and correct any common arithmetic errors.

Board: Project the two problems, then after student work, write final answers with units: (A) E_{net} = compute and direction (show vector signs), (B) $|\tau| = p E \sin\theta$ with numbers and result; note 'no net force in uniform field'.

- Anticipate: Student: For problem A, do we use superposition? — Answer: Yes, add fields from each point charge as vectors.
- Anticipate: Student: For torque, why use $\sin\theta$? — Answer: Because $\tau = |\mathbf{p}| |\mathbf{E}| \sin\theta$ for magnitude from cross product; if $\mathbf{p}$ is perpendicular to $\mathbf{E}$, $\sin\theta = 1$.

Blackboard Notes

- Electric field (point charge)
- Field lines and flux
- Charge densities
- Dipole torque
- Definition: $\mathbf{E}(\mathbf{r})$ is the vector field produced by charges; force on test charge q_0 is $\mathbf{F} = q_0 \mathbf{E}(\mathbf{r})$. Units: $\text{N/C} = \text{V/m}$.
- Point charge: $\mathbf{E}(\mathbf{r}) = (1 / 4\pi\epsilon_0) (Q \hat{\mathbf{r}} / r^2)$. $k = 1 / 4\pi\epsilon_0 \sim 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.
- Field lines: tangent = field direction; density (lines per area perpendicular to field) indicates relative strength; lines start on + and end on -; lines do not cross.
- Flux through area element: $d\Phi = \mathbf{E} \cdot d\mathbf{S}$, where $d\mathbf{S} = \hat{\mathbf{n}} dS$ (units: $\text{N}\cdot\text{m}^2/\text{C}$). If $\mathbf{E}$ perpendicular to area, $d\Phi = E dS$; if at angle θ , $d\Phi = E dS \cos\theta$.
- Charge densities (macroscopic): $\sigma = dQ / dS$ (C/m^2); $\lambda = dQ / dl$ (C/m); $\rho = dQ / dV$ (C/m^3).
- Dipole torque (uniform $\mathbf{E}$): $\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$. Uniform $\mathbf{E} \rightarrow$ torque only (no net force). Non-uniform $\mathbf{E} \rightarrow$ possible net force.
- Common corrections: Number of drawn field lines is arbitrary; the physics is in field strength and line density. Neutral object can be attracted by induced polarization without net charge transfer.
- Diagram: Positive point charge with radial arrows; same picture next to it with twice as many arrows and note 'same field, drawing choice'.
- Diagram: Small rectangular area element with normal $\hat{\mathbf{n}}$ labeled, show angle θ between $\mathbf{E}$ and $\hat{\mathbf{n}}$, label dS and draw $\mathbf{E}$ vector. Annotate $d\Phi = \mathbf{E} \cdot d\mathbf{S}$.
- Diagram: Dipole: $-q$ at $x=-a$, $+q$ at $x=+a$, arrow $\mathbf{p}$ from $-q$ to $+q$. Show uniform $\mathbf{E}$ vector and curved arrow indicating torque.
- Diagram: Sketch of charged rod (comb) near neutral paper bits showing induced polarization (charges separated in the paper) and attraction.
- Formula: $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q \hat{\mathbf{r}}}{r^2}$
- Formula: $\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$

Activities

Quick Field, Flux and Dipole Reasoning (12 min, think_pair_share)

Materials: whiteboard or chart paper, marker for each pair (or one per student), ruler or straightedge, stopwatch

or visible timer, handout with three numbered prompts (one per pair) — teacher can write prompts on board instead

- 1) Arrange students in pairs. Tell groups they will do three short tasks: (A) field from two point charges, (B) field lines and flux through a small patch, (C) dipole in a uniform field. Display or read the three prompts and start a 3-minute individual think time for prompt A.
- 2) After 3 minutes, say “Pair up and explain your answer for A to your partner” and start a 2-minute paired-explain timer. Circulate: check each student writes numbers with units, points to vector directions, and is using vector addition for fields; if a pair is stuck, read the scaffold sentence in Support aloud to one student.
- 3) Repeat the same cycle for prompts B and C: 3 minutes individual thinking, 2 minutes paired explanation for B; then 3 minutes individual thinking, 2 minutes paired explanation for C. While pairs talk, listen to one pair per round and write a short correction on the board if a common error appears (for example, crossing lines or misidentifying start/end of lines).
- 4) Bring the class together for the final 1 minute: call for one quick pair to state their result for each prompt (A, B, C). Correct any persistent errors and explicitly surface the two misconceptions listed: (a) number of drawn field lines is a drawing choice, not intrinsic; (b) induced polarization can produce attraction without net charge transfer.

Student instructions:

- You will work through three short physics prompts. For each prompt: first think alone and write your answer with numbers and units; then explain your reasoning to your partner and listen to theirs.
- Prompt A — Two point charges: $q_1 = +2.0 \mu\text{C}$ at $x = 0$ and $q_2 = -1.0 \mu\text{C}$ at $x = 0.30 \text{ m}$. Find the electric field vector E at the point $x = 0.15 \text{ m}$ (midway). Show steps: compute magnitude of field due to each charge using Coulomb’s law $E = k|q|/r^2$ (use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$), give each field a direction (left or right along x), then add the two vectors and show final magnitude with unit N/C and direction.
- Prompt B — Field lines and flux: On the supplied small square patch (drawn at $x = 0.15 \text{ m}$, side = 0.05 m , normal along $+x$), draw the expected field lines from the two charges above. Then compute the electric flux through the square approximately by treating E as constant across the patch: $\text{flux} = E_x \cdot \text{area}$, where $\text{area} = (0.05 \text{ m})^2$ and E_x is the x -component of the net field at the patch. State units ($\text{N}\cdot\text{m}^2/\text{C}$) and explain whether you would draw more or fewer lines to represent relative strength (and why that does or does not change physical field).
- Prompt C — Dipole torque: Imagine a permanent dipole with dipole moment magnitude $p = 4.0 \times 10^{-29} \text{ C}\cdot\text{m}$ oriented at 60° to a uniform electric field $E = 2.0 \times 10^5 \text{ N/C}$. Derive the torque magnitude by starting from $\tau = pE \sin\theta$ (explain where this comes from as magnitude of p cross E), calculate the numerical value in $\text{N}\cdot\text{m}$, state the direction of torque (tends to rotate dipole to align p with E), and comment whether a net force exists when E is uniform versus when E varies in space.

Success looks like: At least one student in each pair writes numerical values with appropriate units for magnitude of E in Prompt A and points to the directions on their sketch.; Pairs draw field lines for Prompt B that start on $+$ charges and end on $-$ charges and compute a flux value showing area (m^2) times E_x (N/C) giving $\text{N}\cdot\text{m}^2/\text{C}$.; Pairs performing Prompt C write the torque computation showing $\tau = pE \sin\theta$, give a numeric torque in $\text{N}\cdot\text{m}$, and state aloud that net force is zero in a uniform field but nonzero in a non-uniform field.

Support: If you are stuck on Prompt A, use this sentence frame to start: “Field at the mid-point = field from q_1

(magnitude __ N/C to the right/left) plus field from q_2 (magnitude __ N/C to the right/left)." Worked first lines for A: distance from each charge to mid-point = 0.15 m. Compute $E_1 = k \cdot |q_1| / r^2 = (8.99 \times 10^9)(2.0 \times 10^{-6}) / (0.15^2)$ N/C; compute E_2 similarly with $q_2 = 1.0 \times 10^{-6}$ C.

Extension: Harder task: Change q_2 to $+1.0 \mu\text{C}$ (both positive) and place them at $x=0$ and $x=0.30$ m. (a) Find the location(s) on the x -axis where $E = 0$; show algebraic steps and units. (b) For the original signs ($+2.0 \mu\text{C}$ and $-1.0 \mu\text{C}$), estimate the net force on a small dipole ($p = 4.0 \times 10^{-29}$ C·m) placed at $x = 0.15$ m if the field there has a small gradient $dE/dx = 1.0 \times 10^6$ N/(C·m). State whether a net force exists and compute its approximate magnitude using $F \sim (p \cdot dE/dx)$.

Checkpoint Questions

- [apply] Two point charges $q_1 = +4.0 \mu\text{C}$ at $x = 0$ and $q_2 = +1.0 \mu\text{C}$ at $x = 0.08$ m lie on the x -axis. Using Coulomb's law, find the magnitude and direction of the force on q_2 from q_1 . Show units.

Expected: Use $F = (1/4\pi\epsilon_0) q_1 q_2 / r^2$. $r = 0.08$ m. Compute numeric value: $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, so $F = 8.99 \times 10^9 (4.0 \times 10^{-6} \text{ C})(1.0 \times 10^{-6} \text{ C}) / (0.08 \text{ m})^2 = 8.99 \times 10^9 \cdot 4.0 \times 10^{-12} / 0.0064 \sim 5.61$ N. Direction: both charges are positive so q_1 repels q_2 ; q_2 is pushed in $+x$ direction (away from q_1). Answer must state 5.6 N, $+x$ direction (or 'to the right').

- [apply] Three point charges lie on the x -axis: $+2 \mu\text{C}$ at $x = 0$, $-1 \mu\text{C}$ at $x = 0.10$ m, and $+2 \mu\text{C}$ at $x = 0.20$ m. Without heavy arithmetic, explain how you would use superposition to find the electric field at $x = 0.10$ m (the location of the $-1 \mu\text{C}$). State the direction of each contribution and whether the net field at that point is likely to point left or right.

Expected: Superposition: $E_{\text{net}} = E_{\text{from}(+2\mu\text{C at } 0)} + E_{\text{from}(+2\mu\text{C at } 0.20\text{m})}$. At $x = 0.10$ m the left $+2\mu\text{C}$ (at 0) produces a field pointing to the right (away from positive), magnitude $k \cdot 2\mu\text{C} / (0.10 \text{ m})^2$. The right $+2\mu\text{C}$ (at 0.20 m) produces a field pointing to the left, magnitude $k \cdot 2\mu\text{C} / (0.10 \text{ m})^2$. The magnitudes are equal, so the two contributions cancel and $E_{\text{net}} \sim 0$ at $x = 0.10$ m. (A correct answer states directions of each contribution and concludes near zero net field.)

- [understand] A permanent electric dipole with dipole moment $p = 3.0 \times 10^{-6}$ C·m is placed in a uniform electric field $E = (2.0 \times 10^4 \text{ N/C}) \hat{j}$ (i.e., upward). What is the torque on the dipole if p points along $\hat{i}$ (to the right)? Is there a net force on the dipole in this uniform field? Explain briefly.

Expected: Torque $\tau = p \times E$. With $p = (3.0 \times 10^{-6} \text{ C}\cdot\text{m}) \hat{i}$ and $E = (2.0 \times 10^4 \text{ N/C}) \hat{j}$, $\tau = pE (\hat{i} \times \hat{j}) = (3.0 \times 10^{-6})(2.0 \times 10^4) \hat{k} = 6.0 \times 10^{-2} \text{ N}\cdot\text{m} \hat{k}$ (out of the page). State units N·m and direction ($+\hat{k}$ or 'out of page'). Net force in a uniform field is zero because the forces on the positive and negative charges of the dipole are equal and opposite; only a torque remains.

EXIT TICKET

Complete all three parts. (a) Two point charges are on the x -axis: $q_1 = +3.0 \mu\text{C}$ at $x = 0.00$ m and $q_2 = +3.0 \mu\text{C}$ at $x = 0.10$ m. Calculate the magnitude and direction of the electrostatic force on q_2 from q_1 . Show units. (b) A uniform electric field $E = 5.0 \times 10^3 \text{ N/C}$ points in the $+x$ direction. A dipole with moment $p = 2.0 \times 10^{-6} \text{ C}\cdot\text{m}$ is placed so p points along $+x$. Calculate the torque (magnitude and axis/direction) on the dipole and state whether a net force acts on it in that uniform field. (c) Now add a third charge $q_3 = +3.0 \mu\text{C}$ at $x = 0.20$ m. Using superposition, determine the net electric field at $x = 0.05$ m (midpoint between q_1 and q_2). Give magnitude, direction, and units.

Success indicator: Student supplies: (a) numeric F from Coulomb's law with units and direction (about 10 N magnitude, $+x$ direction) computed with k and $r = 0.10$ m; (b) torque $\tau = p \times E$ with numeric value $1.0 \times 10^{-2} \text{ N}\cdot\text{m}$ (or equivalent) and direction ($\hat{k}$ or 'out of page'), and an explicit statement that net force = 0 in a uniform

field; (c) net E at $x=0.05$ m found by adding fields from q_1 and q_2 (and q_3 if it contributes) with numeric magnitude, correct sign/direction along x , and units N/C. Answers must show the use of Coulomb/field formulas and indicate directions.

Homework

- Problem 1 (15 min): Two charges $q_A = +5.0 \mu\text{C}$ at $x = 0$ and $q_B = -2.0 \mu\text{C}$ at $x = 0.12$ m. (a) Compute the force on q_B due to q_A (magnitude and direction). (b) Compute the electric field at $x = 0.06$ m (midway) by superposition; give value and direction in N/C. Show all steps with units.
- Problem 2 (15 min): A dipole with moment $p = 4.0 \times 10^{-6} \text{ C}\cdot\text{m}$ lies at the origin with its axis making 30° above the $+x$ axis in a uniform field $E = 4.0 \times 10^3 \text{ N/C}$ directed along $+x$. (a) Calculate the torque vector (magnitude and direction). (b) Suppose instead the dipole sits near a single positive point charge (non-uniform field). In 2–3 sentences, explain whether the dipole can experience a net force and why (no calculation required).
- Sketching (10 min): Draw field-line diagrams (label directions and explain line-density meaning) for: (i) an isolated positive point charge, (ii) an electric dipole. Under each sketch write one sentence correcting the misconception that 'more drawn lines always means stronger field.'

Estimated time: 40 min

MENTOR MOMENT — COMB, PAPER, AND THE DIPOLE IDEA

In a demo, a teacher combed hair, brought the charged comb near small bits of paper, and a student announced, 'the paper got charged.' The teacher asked how the paper could be attracted without net charge and used a free piece of paper to show one side pulled closer than the other.

Takeaway: A neutral object can be attracted by induction: the external field shifts charges inside the object, creating a dipole that feels a net attraction in a non-uniform field.

Period 6

ENTRY TICKET

You have a very long, uniformly charged straight wire. Without calculation, which of these describes how the electric field magnitude E changes with distance r from the wire?

A) E proportional to $1/r^2$ B) E proportional to $1/r$ C) $E = \text{constant}$ D) E proportional to r

Circle one and write one short sentence explaining your choice (no calculation).

Expected response: Correct choice: B) E proportional to $1/r$. Common incorrect answer: many students pick A) $1/r^2$ because they generalize from point charges. The one-sentence explanation should say whether they thought the wire behaves like a point ($1/r^2$) or that symmetry of a line changes the geometric spreading to $1/r$.

Teacher Script

0-5 min: Collect entry tickets and announce goals

Collect the tickets. Say: "Today we will derive the electric fields for three classic symmetric distributions: an infinite line of charge, an infinite plane sheet, and a thin spherical shell. By the end you should be able to derive each field using Gauss's law and set up an integral for a continuous distribution. Keep your entry ticket answer in mind — we will check it." Pause while collecting, then read one typical wrong answer aloud: "Some of you wrote $1/r^2$ because you remembered the point-charge field — we'll see why a line is different."

Board: Write objectives on the board (short). Underline 'line, plane, shell'.

- Anticipate: Why must the wire be infinite? (To justify translational symmetry and ignore end effects.)
- Anticipate: Is ϵ_0 the same as in Coulomb's law? (Yes; it's the permittivity of free space.)
- Anticipate: What does 'thin spherical shell' mean? (Charge confined to radius R , zero thickness.)

5-15 min: Derive field of an infinite line charge using Gauss's law

Draw a long vertical line on the board and put a small segment labeled ' λ ' (λ = linear charge density, C/m). Say: "We choose a cylindrical Gaussian surface coaxial with the wire. By symmetry the field points radially outward and has the same magnitude at every point on the curved side." Draw a cylinder of radius r and length L around the wire. Ask: "Which parts of the Gaussian surface contribute flux?" Wait for answers: the curved surface only. Then write Gauss's law: $\Phi = Q_{\text{enc}}/\epsilon_0$. Compute $Q_{\text{enc}} = \lambda L$. Compute flux = $E(r) \times (\text{curved area}) = E \times (2\pi r L)$. Rearrange to solve for E : $E = \lambda/(2\pi \epsilon_0 r)$. Say: "Direction: radial ($\hat{n}$).". Point to the entry-ticket misconception: "Notice this gives $1/r$, not $1/r^2$ — because the area of the curved surface grows like r , not r^2 ."

Board: Draw wire and coaxial cylinder. Write: $Q_{\text{enc}} = \lambda L$. Write flux = $E(2\pi r L)$. Write Gauss's law and algebra to $E = \lambda/(2\pi \epsilon_0 r) \hat{n}$.

- Anticipate: Why ignore the flat ends of the cylinder? (Field is parallel to ends; no flux there by symmetry.)
- Anticipate: What if the wire weren't infinite? (You would have end effects; symmetry fails.)
- Anticipate: What is $\hat{n}$? (The outward radial unit vector.)

15-30 min: Derive field of an infinite plane and thin spherical shell (Gauss's law)

First the infinite plane: Draw a large flat sheet and a small Gaussian pillbox that straddles the sheet (height $2h$, faces area A). Say: "By symmetry field is perpendicular to the plane and has the same magnitude on both faces. The sides contribute no flux because field is parallel there."

Write $Q_{\text{enc}} = \sigma A$ ($\sigma = C/m^2$). Flux = $E A + E A = 2 E A$. Use Gauss's law: $2 E A = Q_{\text{enc}}/\epsilon_0 = \sigma A/\epsilon_0$. Solve: $E = \sigma/(2 \epsilon_0)$, direction normal to plane (two directions away from sheet).

Now the thin spherical shell: Draw a sphere of radius R with total charge q on the shell. Explain two regions:

- For $r > R$ choose a spherical Gaussian surface radius r . Symmetry $\Rightarrow$ E radial and constant.

$Q_{\text{enc}} = q$. Flux = $E (4\pi r^2)$. Gauss: $E(4\pi r^2) = q/\epsilon_0 \rightarrow E = q/(4\pi \epsilon_0 r^2)$.

- For $r < R$ choose a Gaussian sphere inside the shell. $Q_{\text{enc}} = 0$ because all charge sits at radius R . So flux = 0 $\Rightarrow E = 0$ inside.

Emphasize units at each step: λ (C/m), σ (C/m²), q (C), ϵ_0 (C²/(N·m²)).

Board: Draw plane with pillbox and label A , h . Write $Q_{\text{enc}} = \sigma A$. Write $2 E A = \sigma A/\epsilon_0 \rightarrow E = \sigma/(2 \epsilon_0)$. Draw sphere with R and inner and outer Gaussian surfaces. Write outer: $E(4\pi r^2) = q/\epsilon_0 \rightarrow E = q/(4\pi \epsilon_0 r^2)$. Write inside:

$Q_{\text{enc}} = 0 \rightarrow E = 0$.

- Anticipate: Why are there two $E A$ terms for the plane? (Because flux passes through both faces of the pillbox with equal magnitude.)
- Anticipate: For the shell, how can field outside be like a point charge? (Because symmetry and Gauss's law show only total charge matters for $r > R$.)
- Anticipate: Does the plane result depend on r ? (No — field is independent of distance for an infinite plane because flux through faces doesn't change with r .)

30-35 min: Quick demonstration of integration idea (superposition)

Say: "Gauss's law used symmetry. The same answers can be reached by integrating Coulomb's law. I'll set up the integral briefly for the line to show the idea." Draw a small line element dl at coordinate along the wire and a point at radial distance r . Write $dQ = \lambda dl$. Write Coulomb's law contribution magnitude $dE = (1/(4\pi \epsilon_0)) dQ / (\text{distance}^2) \times \text{cosine factor}$. Show that integrating around the full infinite wire, the axial components cancel and the radial components add to give E proportional to $1/r$. Do not carry out the full integral (time), but write the integral setup: $E_r = \text{integral } (1/(4\pi \epsilon_0)) (\lambda dl \cos\theta) / s^2$ and note symmetry/geometry reduces it to previous result. Emphasize you can always set up such integrals for less symmetric cases.

Board: Write $dQ = \lambda dl$. Write $dE = (1/(4\pi \epsilon_0)) dQ / s^2$. Write integral form for E_r with a note 'symmetry reduces to previous expression'.

- Anticipate: Why not do the full integral now? (Time; Gauss is quicker for symmetric cases but integration is always possible.)
- Anticipate: What cancels in the integral? (Axial components from elements above and below cancel by symmetry; radial components add.)
- Anticipate: When must we use integration? (When symmetry is insufficient for Gauss's law.)

35-40 min: Check understanding and consolidate

Ask three quick cold-call questions, one per result: (1) "For an infinite line, how does E change with r ?" (expect ' $1/r$ ' and point to formula). (2) "For an infinite plane sheet, what is E and why does it not depend on r ?" (expect ' $\sigma/(2\epsilon_0)$ ', argue from pillbox). (3) "Inside a thin spherical shell is E zero — why?" (expect ' $Q_{\text{enc}} = 0$ inside, so by Gauss $E=0$ '). Write: "If time, copy the boxed formulae and label units." Walk the room while students copy. Collect any final questions and answer briefly.

Board: Box the three final formulae and write units beside each: λ (C/m), σ (C/m²), q (C); write ϵ_0 units C²/(N·m²).

- Anticipate: What if the plane is finite — does E stay constant? (No; edge effects make E vary with distance.)
- Anticipate: How large should L be when using the cylinder? (Large compared to r ; 'infinite' is an idealization.)
- Anticipate: Can I always pick any Gaussian surface? (Yes, but pick one that matches symmetry so E is constant on parts of the surface.)

Blackboard Notes

- Objectives
- Infinite line charge — Gauss
- Infinite plane sheet — Gauss
- Thin spherical shell — Gauss
- Integration setup (line) — superposition
- Objectives: derive E for infinite line (λ), infinite plane (σ), thin spherical shell (q); set up integrals for continuous charge.
- Infinite line: choose cylindrical Gaussian surface coaxial with wire; field radial and constant on curved surface.
- Infinite plane: choose pillbox straddling the plane; field perpendicular and equal on both faces.
- Spherical shell: choose spherical Gaussian surfaces; outside behaves like point charge, inside $E = 0$.
- Integration idea: write dQ for element, apply Coulomb's law, integrate; symmetry simplifies integrals.
- Diagram: Long vertical charged wire with coaxial cylinder (radius r , length L). Label λ , r , L .
- Diagram: Infinite plane with a small pillbox crossing it (area A , half-height h).
- Diagram: Spherical shell radius R with Gaussian spheres $r < R$ and $r > R$.
- Diagram: Small line element dl on wire with point at distance r and angle θ (for integral setup).
- Formula: $E = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$
- Formula: $E = \frac{\sigma}{2\epsilon_0} \hat{n}$
- Formula: $E = \frac{q}{4\pi\epsilon_0 r^2} \hat{r} \quad (r > R)$

Activities

Derive E from Line, Plane and Shell with Gaussian Surfaces (15 min, experiment)

Materials: printed 1-page worksheet per pair (one copy per pair; teacher can print in 5 minutes), whiteboard and marker, ruler (one per pair), pen or pencil (one per student), small round object to represent a spherical shell (ping-pong ball or small rubber ball) and a sheet of paper to represent an infinite plane, a straight object about 20–30 cm long to represent an infinite line (a pen or pencil)

- 1. Arrange students in pairs. Hand each pair the printed worksheet and one ruler, one pen, one small round object, one straight object and one sheet of paper. Say: "In 15 minutes you will derive three electric-field results using Gauss's law and show each Gaussian surface and units." Start the timer for 15 minutes.

•

2. Direct pairs to Task A (infinite line) on the worksheet. Circulate: listen to each pair state their chosen cylindrical Gaussian surface radius r and length L , check they write the expression for flux and for enclosed charge ($\lambda \cdot L$), and verify they carry units (C/m for λ , m^2 for area) through to $E = \lambda / (2\pi \epsilon_0 r)$. If a pair is stuck, point to the cylindrical sketch on the sheet and ask 'which surface pieces contribute flux?'
- 3. After ~5 minutes, ask all pairs to stop and switch to Task B (infinite plane). Repeat circulation: confirm each pair uses a pillbox Gaussian surface, identifies two faces with area A , writes flux = $2EA$ and enclosed charge = $\sigma \cdot A$, and cancels A to get $E = \sigma / (2 \epsilon_0)$ with correct units (C/ m^2 for σ).
- 4. After ~10 minutes total, instruct pairs to do Task C (thin spherical shell). Circulate to confirm students sketch a spherical Gaussian surface both for $r > R$ and $r < R$, compute enclosed charge Q (or 0), and derive $E = Q / (4\pi \epsilon_0 r^2)$ for outside and $E = 0$ for inside, with units. Ask one or two pairs to quickly present their key steps on the whiteboard (one for line or plane, one for shell).
- 5. At the end, collect worksheets or scan a quick photo. Give brief corrective feedback aloud: highlight correct choice of Gaussian surfaces, correct algebra showing cancellations, and explicit units written at each step.

Student instructions:

- You work in pairs. Use the printed worksheet and the classroom objects to draw the Gaussian surface, write the flux integral as $\Phi = \text{closed integral } E \cdot dA$, write the enclosed charge, and derive E step by step. Always write units next to each physical quantity.
- Task A — Infinite line (use the straight object to imagine the charged wire):
 - 1. Sketch a cylinder of radius r and length L coaxial with the line. Label r (m) and L (m).
 - 2. Write the flux through the cylinder: only the curved surface contributes. Flux = $E(r) \cdot (\text{curved area}) = E(r) \cdot (2\pi r L)$. Include units: E in N/C, area in m^2 .
 - 3. Write enclosed charge: $Q_{\text{enclosed}} = \lambda \cdot L$ where λ has units C/m. Substitute into Gauss's law $\Phi = Q_{\text{enclosed}} / \epsilon_0$ and solve for $E(r)$. Show algebra that leads to $E(r) = \lambda / (2\pi \epsilon_0 r)$ and write units for both sides.
- Task B — Infinite plane (use the sheet of paper to imagine the charged plane):
 - 1. Sketch a cylindrical pillbox that crosses the plane, with face area A (m^2). Label normal directions.
 - 2. Write flux: flux = $E_{\text{top}} \cdot A + E_{\text{bottom}} \cdot A = 2 E A$ (both faces). Include units.
 - 3. Write enclosed charge: $Q_{\text{enclosed}} = \sigma \cdot A$ where σ has units C/ m^2 . Use Gauss's law and solve for $E = \sigma / (2 \epsilon_0)$. Show units.
- Task C — Thin spherical shell (use the small ball to represent the shell):
 - 1. Sketch two Gaussian spheres: one with radius $r > R$ (outside), one with $r < R$ (inside). Label R (m), r (m).
 - 2. For $r > R$: write flux = $E \cdot (4\pi r^2)$. Enclosed charge Q = total shell charge Q (C). Use Gauss's law to get $E = Q / (4\pi \epsilon_0 r^2)$. Write units.
 - 3. For $r < R$: state $Q_{\text{enclosed}} = 0$ so flux = 0 and $E = 0$. Explain in one sentence why this follows from symmetry.
- Final small task (on worksheet): for each final formula, state the direction of E with words (radial outward/inward or normal to plane) and list the units of every symbol used (λ , σ , Q , r , ϵ_0).

Success looks like: Each pair has a sketched Gaussian surface for each case (cylinder for line, pillbox for plane, sphere for shell) on their worksheet.; When asked, each pair points to the algebraic step where they substitute

Q_{enclosed} into Gauss's law and cancel common area or length (e.g., cancel A in the plane case, cancel L only where appropriate in line case).; Each pair writes units next to λ , σ , Q , r and shows units on the derived expression (e.g., shows that $\lambda/(2\pi\epsilon_0 r)$ yields N/C).

Support: If you are stuck, start with the sentence frame: 'Choose symmetry $\rightarrow$ Gaussian surface = _____. Only these faces contribute because _____. Flux = $E \cdot (\text{contributing area}) = \text{_____}$. $Q_{\text{enclosed}} = \text{_____}$. Apply Gauss: $E \cdot (\text{area}) = Q_{\text{enclosed}}/\epsilon_0 \rightarrow$ solve for E .' Example worked first item (plane): Choose symmetry $\rightarrow$ plane symmetry, Gaussian surface = pillbox. Only top and bottom faces contribute because E is normal and constant on these faces. Flux = $E \cdot A + E \cdot A = 2EA$. $Q_{\text{enclosed}} = \sigma A$ (σ in C/m^2). Gauss: $2EA = \sigma A/\epsilon_0 \rightarrow$ cancel $A \rightarrow E = \sigma/(2\epsilon_0)$ (units check: $[\sigma] = \text{C/m}^2$, $[\epsilon_0] = \text{C}^2/(\text{N} \cdot \text{m}^2)$ so $[E] = \text{N/C}$).

Extension: Compute the electric field at a point on the perpendicular bisector of a finite straight rod of length $2L$ and uniform linear charge density λ by integrating Coulomb's law. Set up the integral using $dq = \lambda dx$, evaluate E_z at a distance z from the midpoint, carry units through, and show the limiting form as $L \rightarrow \text{infinity}$ reproduces $E = \lambda/(2\pi\epsilon_0 z)$.

Checkpoint Questions

- [apply] You choose a cylindrical Gaussian surface of radius r and length L around an infinitely long line of charge with linear density λ . Write the integral form of Gauss's law for this surface, identify the surfaces that contribute, and solve for $E(r)$.

Expected: Apply Gauss's law: closed integral $E \cdot dA = Q_{\text{enc}}/\epsilon_0$. Only the curved cylindrical surface contributes because end caps are parallel to the field. Curved surface area = $2\pi rL$, enclosed charge $Q_{\text{enc}} = \lambda L$. So $E(2\pi rL) = \lambda L/\epsilon_0 \rightarrow E(r) = \lambda/(2\pi\epsilon_0 r)$ directed radially outward. Units: N/C (or V/m).

- [analyze] Write the integral expression for the electric field at a point a distance r from a short segment dx of an infinite line charge (linear density λ). Show one step that justifies why components along the line cancel, leaving only the radial component to integrate.

Expected: Take a segment dx at position x along the line. dE from Coulomb's law: $dE = (1/(4\pi\epsilon_0)) (\lambda dx)/(r^2 + x^2)$. The component along the line is $dE_{\text{parallel}} = dE \cdot \cos\theta$ with $\cos\theta = x/\sqrt{r^2 + x^2}$. For a symmetric element at $+x$ and $-x$ these parallel components cancel, so only radial component $dE_{\text{radial}} = dE \cdot (r/\sqrt{r^2 + x^2})$ remains. Use $dE_{\text{radial}} = (1/(4\pi\epsilon_0)) (\lambda dx) r/(r^2 + x^2)^{3/2}$ to integrate over x from $-\text{infinity}$ to $+\text{infinity}$.

- [apply] For a uniformly charged thin spherical shell of radius R and total charge Q , explain qualitatively why the electric field is zero everywhere inside the shell ($r < R$) without doing any integrals.

Expected: By symmetry and Gauss's law: choose a spherical Gaussian surface of radius $r < R$ centered on the shell. The enclosed charge is zero because all charge sits on the shell at radius R . Gauss's law gives closed integral $E \cdot dA = Q_{\text{enc}}/\epsilon_0 = 0$, and by symmetry E is the same at every point on that sphere, so $E = 0$ inside. Units: N/C .

EXIT TICKET

An infinite plane sheet carries uniform surface charge density $\sigma = 4.0 \times 10^{-6} \text{ C/m}^2$. Using Gauss's law, find the magnitude and direction of the electric field 3.0 cm on one side of the sheet. Show the Gaussian surface you choose and one sentence that justifies why E is constant on the chosen surface.

Success indicator: Student draws a small pillbox crossing the plane, states flux equation $E(2A) = \sigma A/\epsilon_0$, solves to get $E = \sigma/(2\epsilon_0) = (4.0 \times 10^{-6})/(2\epsilon_0) \sim 2.26 \times 10^3 \text{ N/C}$ and states field is normal to the plane away from the sheet (units N/C). They justify constancy by symmetry: the infinite plane is translationally invariant, so E is the same on

each flat face.

Homework

- Derive the electric field magnitude for an infinitely long line of charge by (a) setting up and evaluating the integral of Coulomb's law for a straight infinite line, and (b) re-deriving the same result using Gauss's law. Write both derivations step by step and show units. (~Hint: both should give $E = \lambda/(2\pi \epsilon_0 r)$).
- Use Gauss's law to show the field of an infinite uniformly charged plane is $E = \sigma/(2\epsilon_0)$. Draw the pillbox Gaussian surface, compute flux, and explain why the two faces contribute equally.
- A thin spherical shell of radius $R = 0.50$ m carries total charge $Q = 2.0$ μC . (a) Calculate the magnitude and direction of the field at $r = 0.30$ m and at $r = 1.0$ m. (b) Briefly explain your result in one sentence each (why zero or why like a point charge).

Estimated time: 45 min

MENTOR MOMENT — WATCH THE SYMMETRY, NOT THE ALGEBRA

A student once integrated the line charge correctly but forgot to argue why axial components cancel, then kept an extra factor of 2 and got a wrong answer on the exam.

Takeaway: Always state the symmetry that makes components cancel before you integrate; it guides the setup and prevents algebraic mistakes.

Learning Gaps & Misconceptions

[high] If the net electric flux through a closed surface is zero, then the electric field must be zero everywhere on (or inside) that surface.

- Diagnostic: A small positive charge $+q$ and an equal negative charge $-q$ sit a short distance apart at the centre of a hollow conducting spherical shell (both charges are inside the hollow, neither touches the shell). The net flux through the spherical shell is zero. Is the electric field at a point on the inner surface of the sphere necessarily zero? Explain your answer.

Reveals: If the student answers that the field must be zero, they are conflating net flux = 0 with local field = 0 and believe cancellation of flux implies no local field anywhere on the surface.

- Diagnostic: A cube sits in a uniform external electric field E_0 directed horizontally. The net flux through the closed cube is zero. Does that imply the electric field vector is zero at points on the cube's faces? Explain.

Reveals: If the student answers the field at points on faces must be zero, they again show they are equating net (signed) flux cancellation with a zero local field instead of recognising opposite faces contribute equal and opposite flux while local field can be nonzero.

- Remediation: Show a diagram (or use a simulation) of an electric dipole inside a transparent sphere. Have students draw a few field lines from the positive to negative charge and then mark through which small surface patches those lines enter and leave the sphere. Ask them to compute the signed flux contribution of a chosen patch ($\Phi = E \cdot A$, with units $N \cdot m^2/C$) and then add several patch contributions to show how positive and negative terms cancel to give net zero flux while E on patches is nonzero.

Why: Concrete drawing/calculation ties the abstract scalar flux to local vector contributions. Carrying units and computing $\Phi = E \cdot A$ for patches forces students to see flux is a sum of signed local quantities; cancellation of those signed numbers can yield zero net flux even though the magnitude of E at patches is large. This attacks the false generalisation from symmetry and the 'zero means nothing there' intuition.

- Remediation: Run a quick hands-on or virtual activity with a uniform field through a cube: pick one face, measure (or compute) $E \cdot A$ on each face (show signs explicitly) and sum them. Then ask: 'Does any individual face have zero field?' and have students justify with units and vector dot product why faces have nonzero flux magnitude but opposite sign.

Why: Using a simple uniform-field example isolates the mechanism of opposite-signed contributions (incoming vs outgoing) and shows algebraically why the net is zero while local E is not. Requiring dot products and units reinforces the distinction between vector field values (N/C) and scalar flux ($N \cdot m^2/C$).

- Remediation: Give a short guided derivation: start from the definition $\Phi = \text{closed integral } E \cdot dA$ and expand for a surface partitioned into small elements. Lead students to the conclusion that Gauss's law says closed integral $E \cdot dA = Q_{\text{enclosed}}/\epsilon_0$ but does not say anything about the value of E at a given element. Finish with a quick problem: place two equal charges inside a sphere and ask for (a) total flux, (b) E at a chosen point on the surface — students must compute or estimate both.

Why: Deriving the integral form step by step, with explicit units and small-area elements, prevents students from accepting Gauss's law as a black-box rule and clarifies what the law constrains (the integral) versus what it does not (pointwise E). The guided problem forces application of that distinction.

[high] The bigger charge pushes harder, so the larger charge exerts a bigger force on the smaller charge than the smaller charge exerts on the larger one.

- Diagnostic: Two isolated point charges, $+5\ \mu\text{C}$ and $+1\ \mu\text{C}$, are fixed $0.50\ \text{m}$ apart. Which of these statements is correct about the magnitudes of the electrostatic forces between them? A) The $+5\ \mu\text{C}$ charge exerts a force five times as large on the $+1\ \mu\text{C}$ charge as the $+1\ \mu\text{C}$ charge exerts on the $+5\ \mu\text{C}$. B) Each charge experiences the same magnitude of force, but in opposite directions. C) The forces depend on the masses, so the lighter one feels a larger force.

Reveals: If a student picks A or C, they believe forces are asymmetric and that the larger charge 'produces' a bigger force on the smaller charge (or that mass changes the force), showing they haven't internalised the mutual nature of Coulomb force.

- Diagnostic: You measure that the force on charge A (due to charge B) is $2.0\ \text{N}$ to the right. What is the force on charge B (due to charge A)?

Reveals: A wrong answer that gives a different magnitude (e.g. 'less than $2.0\ \text{N}$ ' or 'zero if B is small') shows the student thinks the interaction is one-way or depends on which is 'acting' rather than being mutual and equal in magnitude.

- Remediation: Start with the algebraic form $F = k q_1 q_2 / r^2$ written for the force on charge 1 due to charge 2 and then explicitly write the force on charge 2 due to charge 1, showing the same expression. Do a step-by-step numeric example with $q_1 = +5\ \mu\text{C}$ and $q_2 = +1\ \mu\text{C}$ at $r = 0.50\ \text{m}$, carrying units through the calculation and producing the numeric force for both 'force on 1 due to 2' and 'force on 2 due to 1'.

Why: Deriving both expressions side-by-side ties the abstract formula to concrete numbers and units and shows algebraically that the magnitudes are identical; carrying units and plugging numbers prevents students from treating the formula as a black box and confronts the incorrect intuition with a direct calculation.

- Remediation: Use a classroom demo or simulation with two charged pith-ball spheres or an online electrostatics simulation that displays force vectors and magnitudes on each charge. Arrange unequal charges, display the equal-length force vectors acting in opposite directions, and let students change which charge is labelled 'source' and 'test' while observing that the displayed magnitudes remain equal.

Why: A visual, measurable representation confronts the everyday intuition: when students see equal-length arrows on both objects despite unequal charges, they reconcile the algebra with a physical picture. Allowing them to swap labels addresses the mistaken idea that a 'source' versus 'test' role changes the force.

- Remediation: Pose a short reasoning task: ask students to predict what would happen to the total momentum of an isolated two-charge system when they exert mutual electrostatic forces, and connect this to Newton's third law and conservation of momentum. Work through a quick derivation showing that equal-and-opposite forces are required for internal forces to conserve total momentum (use impulse $\Delta p = F\Delta t$).

Why: Linking Coulomb's law symmetry to Newton's third law and to conservation of momentum gives a physical conservation principle that explains why forces must be equal and opposite, addressing the root cause (everyday asymmetric intuition) with a fundamental symmetry argument rather than a bare assertion.

[medium] When two charged objects touch, their charges are destroyed — the charge just disappears instead of being shared or moved to other objects.

- **Diagnostic:** You have a small metal sphere A carrying $+4.0 \times 10^{-6} \text{ C}$ and an identical isolated sphere B carrying $-1.5 \times 10^{-6} \text{ C}$. You touch them together and then separate them. What is the charge on each sphere after separation?

Reveals: If the student answers with a smaller magnitude or zero total and does not conserve algebraic charge, they treat contact as destroying charge rather than redistributing it.

- **Diagnostic:** A positively charged rod touches a neutral metal can. After touching, the can is less positive or appears uncharged. Does this mean the positive charge on the rod was destroyed? Explain briefly.

Reveals: A student saying the charge was 'destroyed' or 'used up' demonstrates the misconception; a correct answer should say the charge moved to other parts of the system or was locally neutralised by electrons transferred from the can while total charge is conserved.

- **Remediation: Demonstration and measurement:** Take two conductive spheres on insulating stands connected to a coulomb meter or electroscope calibrated in coulombs. Charge sphere A to $+6.0 \times 10^{-6} \text{ C}$ (measure and record), charge sphere B to $-2.0 \times 10^{-6} \text{ C}$ (measure and record). Bring them into contact, then separate them. Measure and record the charge on each sphere after separation and calculate the algebraic sum before and after.

Why: Concrete measurements force students to track signed charge (with units coulombs) and see numerically that the algebraic sum before and after contact is the same, showing charge is redistributed, not destroyed. Using measured values prevents reliance on vague language like 'neutralised' and anchors abstract conservation in data.

- **Remediation: Thought-experiment with isolation:** Ask students to imagine the two spheres and the meter form a closed system. Guide them through writing the initial total charge (including signs), then show via stepwise charge-transfer reasoning or simple algebra that after contact the total algebraic charge must be unchanged because no charge left the isolated setup. Have students predict the final individual charges (equal sharing for identical conductors) and compare with the measured result.

Why: Explicitly framing an isolated system and carrying algebraic signs step by step addresses the reasoning error: it replaces the vague idea of 'lost' charge with the conservation argument and shows how local neutralisation is cancellation of opposite signs within the system while global charge remains constant.

- **Remediation: Counterexample with external conductor:** Repeat a short demonstration where one charged object touches a large grounded conductor (earth). Measure the charge of the object before and after, and measure/argue where charge went (to the ground). Ask: was charge destroyed? Have students contrast this with the isolated-spheres case and write a short explanation in units (C) of what happened to the algebraic total.

Why: Showing a case where charge truly leaves the closed system (to ground) clarifies the difference between disappearance from a local object and destruction. It reinforces that 'charge gone' usually means 'moved elsewhere' unless current flows out of the system — making conservation explicit and tying the concept to measurable transfer with units.

[medium] If a diagram shows more field lines coming out of one charge than another, that charge really 'has' more field lines and therefore a stronger electric field there; the absolute count of drawn lines is a physical quantity.

- Diagnostic: Two identical isolated positive point charges Q are drawn on separate diagrams. Diagram A shows 12 radial field lines drawn from its surface; Diagram B shows 24 radial field lines from the other charge. Which charge produces the stronger electric field at a point 1.0 m from its center?

Reveals: Whether the student treats the absolute number of drawn lines as a physical property rather than recognising that the diagrams use different numbers of lines arbitrarily; a wrong answer reveals they equate 'more lines drawn' with stronger actual field.

- Diagnostic: You are shown two diagrams of the same point charge. In diagram X the artist has drawn 8 field lines; in diagram Y the artist has drawn 16 field lines, but both diagrams use the same spacing so the lines in Y are thinner and the distance between adjacent lines is the same as in X. At a location 0.5 m from the charge, does the electric field magnitude differ between diagram X and diagram Y? Explain briefly.

Reveals: Whether the student recognises that the drawing conventions (number of lines) do not change the physical field and that only line density per unit area matters; a wrong answer shows they count lines rather than assess local density or the underlying formula for E .

- Remediation: Start class with a paired-diagram demonstration: draw two identical point-charge field diagrams on the board. For Charge A draw 8 evenly spaced radial lines; for Charge B draw 24 radial lines but make the area over which they spread three times larger so that the local density (lines per unit arc length at a given radius) is the same as A. Ask students to predict which has larger E at a fixed radius (e.g. $r = 1.0$ m) and then measure with a small test charge on a calibrated force spring scale attached to a long insulating rod (or calculate from Coulomb's law) to show the same force in both cases.

Why: The side-by-side visual plus a physical measurement ties the abstract concept of field magnitude to a measurable force on a test charge and shows that changing the absolute number of drawn lines (an arbitrary artist choice) does not change the physical E . Making local line density equal isolates the drawing choice as the only difference.

- Remediation: Have students redraw a single point-charge field three ways: (a) few thick lines, (b) many thin lines packed so density is the same as (a), and (c) many lines but spread farther apart so local density is lower. For each drawing ask them to compute or estimate the field magnitude at a fixed radius using Coulomb's law $E = kQ/r^2$ (explicitly carrying units) and then annotate each drawing with 'relative field strength = ... N/C '.

Why: Requiring students to produce multiple drawings and then calculate E links the pictorial density to the quantitative formula. Explicit units and numbers prevent vague appeals to 'more lines' and force them to reason: identical Q and r give identical E regardless of line count; only density correlates with E when drawings are consistent in area.

- Remediation: Introduce a short class activity comparing 'lines per unit area' to flux: place a small circular ring at radius r around a drawn charge and count how many field lines cross the ring in different drawings. Ask students to relate that count per unit ring length to the calculated field magnitude and to explain why doubling all drawn lines everywhere does not change lines per unit area if spacing scales accordingly.

Why: Counting lines through a fixed small area makes the abstract idea of local line density concrete and

connects directly to the physical idea of flux or field strength through an area. Demonstrating that uniformly scaling the total number of drawn lines (without changing density) leaves crossings per area unchanged addresses the root confusion that absolute counts are intrinsic.

[medium] If a neutral object (like a small piece of paper) is attracted to a charged object (like a rubbed comb), that paper must have become net charged — otherwise how could it feel a force?

- **Diagnostic:** A plastic comb is rubbed on wool and held near small neutral bits of paper on a table. The paper jumps up and sticks to the comb without touching it. Has the paper gained net charge? Explain your answer briefly.

Reveals: If the student says the paper must have gained net charge, they are treating attraction as always requiring net charging and are not recognising induced polarisation or the role of field non-uniformity.

- **Diagnostic:** Two different experiments: (A) a charged rod held near tiny neutral metal filings causes them to move toward and cling to the rod but if you touch them they jump away; (B) a charged rod placed near a neutral insulating bead fixed in place causes the bead to shift toward the rod but it remains electrically neutral when measured afterward. Must both objects have net charge in each case?

Reveals: If the student asserts that both objects must be net charged, they are not distinguishing induced dipole attraction (no net charge needed) from cases where charge transfer occurs (e.g., conduction when touching).

- **Remediation: Demonstration:** bring a charged plastic comb near lightweight neutral paper bits and observe attraction. Then repeat with a metallic foil strip connected to an electroscope so the foil can be checked for net charge without touching the comb. Show that the paper/foil is attracted but the electroscope shows no net charge unless contact occurs.

Why: Directly ties the observable attraction to a measurement of net charge. Students see attraction happen while an instrument confirms the object remains neutral, forcing them to reconcile motion with no net charge and opening discussion of induced charge separation rather than charge transfer.

- **Remediation: Guided reasoning with field sketches and force comparisons:** draw a charged rod and a neutral atom/insulator near it. Have students sketch induced charge separation (negative side nearer if rod positive), then compute qualitatively the forces on nearer and farther charges using $1/r^2$ dependence. Ask: are the forces equal in magnitude? Which is stronger and why? Conclude net force direction.

Why: Shows the cause of net attraction quantitatively grounded in the field law students know (inverse square). They must derive the net force from separate contributions rather than accept a statement — this addresses the overgeneralisation by using a narrower true principle (distance dependence) to explain the effect.

- **Remediation: Contrast activity:** small group tasks comparing conduction vs induction. Give scenarios (touching vs no touching, conductor vs insulator, uniform vs non-uniform external field) and ask students to predict whether the object ends up with net charge and whether it will be attracted. Then perform quick experiments or simulations for each scenario.

Why: Helps students form a decision tree linking observable conditions to outcomes (net charge vs induced dipole) so they stop automatically concluding net charging. By varying conductors/insulators and contact, they see when charge transfer is necessary and when polarisation suffices.

[medium] Linear/surface/volume density just means Q divided by the whole length/area/volume, so to get the charge on any piece I can always do $Q_{\text{piece}} = (\text{total } Q / \text{total length}) \times \text{piece length}$ (or the analogous area/volume fraction) — density is a single constant number you can use on any scale.

- Diagnostic: A 1.0 m rod carries total charge $Q = 6.0 \mu\text{C}$ and its linear charge density varies with position as $\lambda(x) = 2.0 \times 10^{-6} \text{ C/m} \times x$, where x is measured from one end ($0 \leq x \leq 1.0 \text{ m}$). What is the charge on the left half of the rod ($0 \leq x \leq 0.5 \text{ m}$)?

Reveals: If the student answers by taking the fraction of length times the total charge, they are treating density as a constant global average and not integrating the position-dependent $\lambda(x)$.

- Diagnostic: A circular disk radius R has surface charge density $\sigma(r) = \sigma_0 (r/R)$ (it increases linearly from centre to edge). Is the charge on an annulus from $r = 0.4R$ to $r = 0.6R$ equal to the fraction of total charge given by (area of that annulus)/(area of disk) $\times Q_{\text{total}}$? Explain and give a numerical comparison using σ_0 and R .

Reveals: If the student insists the annulus carries exactly that area fraction of Q_{total} , they are again assuming a uniform density and failing to account that σ depends on r and must be integrated over the area with the correct area element.

- Remediation: Derive $dq = \lambda(x) dx$ (and the analogous σdA , ρdV) from the limit definition in class: start with partitioning a non-uniform rod into N equal-length pieces, write $Q \sim \sum \lambda(x_i) \Delta x$, show units at every step ($\text{C/m} \times \text{m} = \text{C}$), then take the limit $N \rightarrow \infty$ to obtain the integral expression $Q = \int \lambda(x) dx$. Do this with a concrete $\lambda(x) = kx$ and numbers so students compute both the Riemann sum for small N and the integral.

Why: Walking through partition $\rightarrow$ sum $\rightarrow$ limit confronts the mistaken shortcut directly and makes clear that λ in the integrand is a local (position-dependent) value evaluated on each infinitesimal piece; carrying units ties the abstract symbol to measurable charge. The numeric Riemann-sum vs integral comparison shows the failure of the 'area-fraction' shortcut for non-uniform λ .

- Remediation: Hands-on worksheet activity: give students a printed rod image with $\lambda(x) = kx$ (with k and rod length numeric), ask them to estimate charge on subinterval $[0, L/2]$ by (a) treating λ as constant equal to the average (Q_{total}/L), (b) taking λ at the midpoint times length, and (c) evaluating the integral $\int_0^{L/2} \lambda(x) dx$. Have them compute percent error of (a) and (b) versus (c) and discuss.

Why: Explicitly comparing the incorrect shortcut calculations to the correct integral calculation and seeing large errors for (a)/(b) drives home that density can vary and that only the integral gives the correct charge; the activity makes the abstract concept concrete and computationally checkable.

- Remediation: Visualise surface/volume densities with a classroom demonstration: project two images of a disk — one uniformly shaded and one with shading increasing with radius — and ask students to predict which annulus carries more charge. Then, derive and compute the charge on equal-area annuli for both cases (perform the integral for the graded disk). Emphasise units ($\text{C/m}^2 \times \text{m}^2 = \text{C}$) and show numeric results.

Why: A visual comparison links intuition to the mathematics: students see that identical-area regions can carry different charge if σ varies. Performing the integral immediately after the visual reinforces that local $\sigma(r)$ times the local area element must be summed (integrated) to find charge, correcting the area-fraction misconception.

[medium] If a third charge is put near two charges, it changes the force that one of those two charges exerts on the other — you can't just add forces from each charge; the pairwise interaction itself is altered by the presence of others.

- Diagnostic: Charge A and charge B are fixed in space. You measure the electrostatic force that A exerts on B. Now place a third charge C near B (but not touching) and keep A and B fixed. Does the electrostatic force that A exerts on B change in magnitude or direction because of C? Explain your answer.

Reveals: A wrong answer shows the student thinks the presence of C alters the pairwise force between A and B (confusing net force with pairwise interaction or believing interactions are non-additive).

- Diagnostic: Three point charges lie on a line: $+2Q$ at $x = -d$, $+Q$ at $x = 0$ (middle), and $-Q$ at $x = +d$. Calculate the electrostatic force on the middle charge by summing contributions. A student instead replaces the two outer charges by a single net charge $(+2Q + (-Q) = +Q)$ placed at $x = 0$ and then says 'net force on the middle charge is zero'. Is that correct? If not, calculate the correct net force on the middle charge and say why the student's replacement is invalid.

Reveals: A wrong answer shows the student believes you can replace spatially separated charges by their algebraic sum at a single location (ignores vector nature and distance dependence), i.e., they do not apply superposition correctly.

- Remediation: Demonstration/simulation: place two fixed charges A and B on-screen or with a Coulomb kit and measure (or calculate) the force on B due to A. Then add a third charge C near B; keep A and B fixed and separately calculate/measure the force on B due to A alone, due to C alone, and the net force (vector sum). Display the algebraic expressions for F_{AB} and F_{CB} and show numerically that F_{AB} is unchanged by C.

Why: Seeing the measured/calculated F_{AB} remain equal before and after adding C, while the net force changes only because of an extra term, confronts the false belief that the pairwise interaction itself changes. Writing F_{AB} explicitly shows the pairwise Coulomb expression contains only A and B variables, addressing the root confusion.

- Remediation: Guided derivation and worked example: derive net force on a test charge q at position r_0 due to two charges q_1 at r_1 and q_2 at r_2 . Write $F_{\text{total}} = k q [(q_1 (r_0 - r_1) / |r_0 - r_1|^3) + (q_2 (r_0 - r_2) / |r_0 - r_2|^3)]$. Have students identify the separate terms and point out which letters (positions/charges) appear in each term, then vary q_2 and show algebraically that the term for q_1 does not change.

Why: A step-by-step derivation ties the abstract formula to positions and units and makes explicit that each pair contributes an independent vector term. This addresses the source error (students not seeing the algebra that proves superposition) rather than just asserting the rule.

- Remediation: Counter-analogy and sorting exercise: give students several short scenarios (e.g., people pushing a box from different sides, charges at different locations) and ask them to list whether you can 'combine' sources by adding magnitudes or must add vectors after accounting for location. Include one where two equal charges are on opposite sides of a test charge so students must compute two opposite forces to see cancellation depends on distance and direction, not just net algebraic charge.

Why: The analogy workshop forces them to distinguish when simple scalar addition is valid and when spatial arrangement and vector direction matter. Sorting concrete cases builds intuition that you cannot collapse spatially separated sources to a single equivalent at a different location, addressing the everyday-experience

generalisation.